

Algebraic structures. Cohomology of groups

SGA 3, Exposé I

English draft

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About this draft

This is an unofficial English translation of Exposé I, “Algebraic structures. Cohomology of groups” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp1-13oct24.pdf` (48 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo: comment`; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:.` Scholarly proofreading remains outstanding.

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Contents

About this draft	iii
Exposé I. Algebraic structures. Cohomology of groups	1
1. Generalities	1
2. Algebraic structures	9
3. The category of $\mathbf{O}$ -modules, the category of $\mathbf{G-O}$ -modules	14
4. Algebraic structures in the category of schemes	16
4.4. Diagonalizable groups	18
5. Cohomology of groups	24
6. G -equivariant objects and modules	30

Exposé I. Algebraic structures. Cohomology of groups

by M. Demazure

⁽⁰⁾This exposé consists of two parts; the first collects a certain number of general definitions and sets down notations which will often be used in the sequel, the second treats the cohomology of groups and leads to Theorem 5.3.3 (vanishing of the cohomology of diagonalizable groups).

We choose once and for all a Universe.⁽¹⁾ All the definitions introduced and all the constructions carried out will be relative to this Universe. We shall systematically allow ourselves the following abuse of language: in order to define a functor $f: \mathcal{C} \rightarrow \mathcal{C}'$, we shall content ourselves with defining the object $f(S)$ of $\mathcal{C}'$ for every object S of $\mathcal{C}$, whenever there will be no ambiguity as to the manner of defining $f(h)$ for an arrow h of $\mathcal{C}$. In practice, we shall say: let $f: \mathcal{C} \rightarrow \mathcal{C}'$ be the functor defined by $f(S) = \dots$.

1. Generalities

1.1. — Let $\mathcal{C}$ be a category. One will denote by $\widehat{\mathcal{C}}$ the category $\text{Hom}(\mathcal{C}^\circ, (\text{Ens}))$ of contravariant functors from $\mathcal{C}$ to the category (Ens) of sets.⁽²⁾ There exists a canonical functor $\mathbf{h}: \mathcal{C} \rightarrow \widehat{\mathcal{C}}$ which associates with every $X \in \text{Ob}(\mathcal{C})$ the functor $\mathbf{h}_X$ such that

$$\mathbf{h}_X(S) = \text{Hom}(S, X).$$

For every functor $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$, one defines (cf. for example EGA 0_{III}, 8.1.4) a bijection⁽³⁾

$$\text{Hom}(\mathbf{h}_X, \mathbf{F}) \xrightarrow{\sim} \mathbf{F}(X).$$

In particular, for every pair X, Y of objects of $\mathcal{C}$, the canonical map below is bijective:

$$\text{Hom}_{\mathcal{C}}(X, Y) \xrightarrow{\sim} \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{h}_X, \mathbf{h}_Y);$$

i.e. the functor $\mathbf{h}$ is *fully faithful*. It therefore defines an isomorphism of $\mathcal{C}$ onto a *full* subcategory of $\widehat{\mathcal{C}}$, and an equivalence of $\mathcal{C}$ with the *full* subcategory of $\widehat{\mathcal{C}}$ formed of the *representable* functors (i.e. isomorphic to a functor of the form $\mathbf{h}_X$). In the sequel, we shall often identify X and $\mathbf{h}_X$. The following numbers are intended to show that this identification can be made without danger.

Remark 1.1.1. ⁽⁴⁾One sometimes needs the following variant. Let $\mathcal{D}$ be a *full* subcategory of $\mathcal{C}$ and let $X, Y \in \text{Ob}(\mathcal{D})$; denote by $\mathbf{h}'_X$ and $\mathbf{h}'_Y$ the restrictions to $\mathcal{D}$ of $\mathbf{h}_X$ and $\mathbf{h}_Y$. Then one has

$$\text{Hom}_{\mathcal{C}}(X, Y) = \text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_{\widehat{\mathcal{D}}}(\mathbf{h}'_X, \mathbf{h}'_Y)$$

and hence: to give a morphism $X \rightarrow Y$ “is the same thing” as to give, in a manner functorial in T , a map $\varphi(T): \text{Hom}(T, X) \rightarrow \text{Hom}(T, Y)$, for every $T \in \text{Ob}(\mathcal{D})$.

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** Cf. SGA 4, Exp. I, §0 and the Appendix; see also the discussion in [DG70], p. xxvi.

⁽²⁾**N.D.E.** One calls it the category of presheaves on $\mathcal{C}$, cf. IV 4.3.1.

⁽³⁾**N.D.E.** This result is often called the “Yoneda lemma”; we shall use this terminology in other N.D.E.

⁽⁴⁾**N.D.E.** This remark has been added.

1.2. — ⁽⁵⁾One will say that $\mathbf{F}$ is a *subobject* (or a *subfunctor*) of $\mathbf{G}$ if $\mathbf{F}(S)$ is a subset of $\mathbf{G}(S)$ for each S .

In $\widehat{\mathcal{C}}$ “arbitrary” projective limits exist and are computed by:⁽⁶⁾

$$(\varprojlim_i \mathbf{F}_i)(S) = \varprojlim_i \mathbf{F}_i(S).$$

In particular, fibered products are defined by:⁽⁷⁾

$$(\mathbf{F} \times_{\mathbf{G}} \mathbf{F}')(S) = \mathbf{F}(S) \times_{\mathbf{G}(S)} \mathbf{F}'(S).$$

We shall choose as final object of $\widehat{\mathcal{C}}$ the functor $\mathbf{e}$ such that $\mathbf{e}(S) = \{\emptyset\}$.⁽⁸⁾ Every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$ possesses a unique morphism into $\mathbf{e}$, and one sets

$$\mathbf{F} \times \mathbf{F}' = \mathbf{F} \times_{\mathbf{e}} \mathbf{F}'.$$

The functor $\mathbf{h}$ commutes with projective limits; in particular, for $X \times X'$ to exist ($X, X' \in \text{Ob}(\mathcal{C})$), resp. for $\mathcal{C}$ to admit a final object e , it is necessary and sufficient that $\mathbf{h}_X \times \mathbf{h}_{X'}$ be representable, resp. that $\mathbf{e}$ be representable, and one has

$$\mathbf{h}_X \times \mathbf{h}_{X'} \simeq \mathbf{h}_{X \times X'} \quad \text{and} \quad \mathbf{h}_e \simeq \mathbf{e}.$$

A monomorphism of $\widehat{\mathcal{C}}$ is nothing other than a morphism $\mathbf{F} \rightarrow \mathbf{G}$ such that, for $S \in \text{Ob}(\mathcal{C})$, the corresponding map of sets $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is injective.⁽⁹⁾

The functor Γ . For every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$ one sets

$$\Gamma(\mathbf{F}) = \text{Hom}(\mathbf{e}, \mathbf{F});$$

an element of $\Gamma(\mathbf{F})$ is thus a family $(\gamma_S)_{S \in \text{Ob}(\mathcal{C})}$, $\gamma_S \in \mathbf{F}(S)$, such that for every arrow $f: S' \rightarrow S''$ of $\mathcal{C}$, one has $\mathbf{F}(f)(\gamma_{S''}) = \gamma_{S'}$.

One sets $\Gamma(X) = \Gamma(\mathbf{h}_X)$ for $X \in \text{Ob}(\mathcal{C})$. If $\mathcal{C}$ has a final object e , one thus has an isomorphism $\Gamma(X) \simeq \text{Hom}(e, X)$.

1.3. — Let $S \in \text{Ob}(\mathcal{C})$. One denotes by $\mathcal{C}/S$ the category of objects of $\mathcal{C}$ over S , i.e. the category whose objects are the arrows $f: T \rightarrow S$ of $\mathcal{C}$, the set $\text{Hom}(f, f')$ being the subset of $\text{Hom}(T, T')$ formed of the u such that $f = f' \circ u$. If $\mathcal{C}$ possesses a final object e , then $\mathcal{C}/e$ is isomorphic to $\mathcal{C}$. The category $\mathcal{C}/S$ possesses a final object: the identity arrow $S \rightarrow S$.

If $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, then one can form the category $(\mathcal{C}/S)/f$, which one denotes, by abuse of language, $(\mathcal{C}/S)/T$, and one has a canonical isomorphism

$$\mathcal{C}/T \simeq (\mathcal{C}/S)/T.$$

This construction also applies to the category $\widehat{\mathcal{C}}$; one defines in particular the category $\widehat{\mathcal{C}}/\mathbf{h}_S$. On the other hand, one can form the category $\widehat{\mathcal{C}}/S$.

If $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, then $\Gamma(f)$ is identified with the set $\Gamma(T/S)$ of sections of T over S , that is, of the arrows $S \rightarrow T$ which are right inverses of f . Let us observe that $\mathbf{h}_f: \mathbf{h}_T \rightarrow \mathbf{h}_S$ is then an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$ and that one has:

$$\Gamma(\mathbf{h}_f) \simeq \Gamma(\mathbf{h}_T/\mathbf{h}_S) \simeq \Gamma(T/S) \simeq \Gamma(f).$$

⁽⁵⁾**N.D.E.** The order has been modified, so as to introduce fibered products before monomorphisms, cf. N.D.E. (9).

⁽⁶⁾**N.D.E.** Likewise, “arbitrary” inductive limits exist and are computed “argument by argument”, i.e. $(\varinjlim_i \mathbf{F}_i)(S) = \varinjlim_i \mathbf{F}_i(S)$; but in general the functor $\mathbf{h}$ does not commute with inductive limits.

⁽⁷⁾**N.D.E.** In particular, the kernel of a pair of morphisms $u, v: \mathbf{F} \rightrightarrows \mathbf{G}$ is the subfunctor $\text{Ker}(u, v)$ of $\mathbf{F}$ defined by $\text{Ker}(u, v)(S) = \{x \in \mathbf{F}(S) \mid u(x) = v(x)\}$.

⁽⁸⁾**N.D.E.** $\{\emptyset\}$ (the set of subsets of the empty set) denotes the one-element set.

⁽⁹⁾**N.D.E.** If $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is injective for every S , it is clear that $\mathbf{F} \rightarrow \mathbf{G}$ is a monomorphism; the converse is seen by considering the diagram $\mathbf{F} \times_{\mathbf{G}} \mathbf{F} \rightrightarrows \mathbf{F} \rightarrow \mathbf{G}$. One thus obtains that: “ $\mathbf{F} \rightarrow \mathbf{G}$ is a monomorphism if and only if the diagonal morphism $\mathbf{F} \rightarrow \mathbf{F} \times_{\mathbf{G}} \mathbf{F}$ is an isomorphism” (cf. EGA I, 5.3.8). Likewise, it is clear that if $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is surjective for every S , then $\mathbf{F} \rightarrow \mathbf{G}$ is an *epimorphism*, and the converse is seen by considering the amalgamated sum $\mathbf{G} \amalg_{\mathbf{F}} \mathbf{G}$, cf. the proof of Lemma 4.4.4 in Exp. IV.

1.4. — One now proposes to define an equivalence of the categories $\widehat{\mathcal{C}/S}$ and $\widehat{\mathcal{C}}/\mathbf{h}_S$, that is, to prove that “to give a functor on the category of objects of $\mathcal{C}$ over S , is “the same thing” as to give a functor on $\mathcal{C}$ equipped with a morphism into $\mathbf{h}_S$ ”.

(i) *Construction of $\alpha_S: \widehat{\mathcal{C}}/\mathbf{h}_S \rightarrow \widehat{\mathcal{C}/S}$.*

Let first $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ be an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$. One must define a functor $\alpha_S(\mathbf{H})$ on $\mathcal{C}/S$. Let therefore first $f: T \rightarrow S$ be an object of $\mathcal{C}/S$; let us define $\alpha_S(\mathbf{H})(f)$ to be the inverse image of $f \in \mathbf{h}_S(T)$ under the map $\mathbf{H}(T): \mathbf{F}(T) \rightarrow \mathbf{h}_S(T)$.⁽¹⁰⁾

Let next $u: f \rightarrow f'$ be an arrow of $\mathcal{C}/S$; then $\mathbf{F}(u): \mathbf{F}(T') \rightarrow \mathbf{F}(T)$ induces a map from $\alpha_S(\mathbf{H})(f')$ to $\alpha_S(\mathbf{H})(f)$, which one denotes $\alpha_S(\mathbf{H})(u)$. One checks at once that the maps

$$f \mapsto \alpha_S(\mathbf{H})(f) \quad \text{and} \quad u \mapsto \alpha_S(\mathbf{H})(u)$$

do define a functor on $\mathcal{C}/S$, hence an object $\alpha_S(\mathbf{H})$ of $\widehat{\mathcal{C}/S}$.

Let finally $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ and $\mathbf{H}': \mathbf{F}' \rightarrow \mathbf{h}_S$ be two objects of $\widehat{\mathcal{C}}/\mathbf{h}_S$ and $\mathbf{U}: \mathbf{F} \rightarrow \mathbf{F}'$ a morphism of $\widehat{\mathcal{C}}/\mathbf{h}_S$:

$$\begin{array}{ccc} \mathbf{F} & \xrightarrow{\mathbf{U}} & \mathbf{F}' \\ & \searrow \mathbf{H} & \swarrow \mathbf{H}' \\ & \mathbf{h}_S & \end{array}$$

Then for every $f: T \rightarrow S$, the map $\mathbf{U}(T): \mathbf{F}(T) \rightarrow \mathbf{F}'(T)$ induces a map

$$\alpha_S(\mathbf{U})(f): \alpha_S(\mathbf{H})(f) \rightarrow \alpha_S(\mathbf{H}')(f),$$

which defines a morphism of functors

$$\alpha_S(\mathbf{U}): \alpha_S(\mathbf{H}) \rightarrow \alpha_S(\mathbf{H}').$$

One checks easily that the maps

$$\mathbf{H} \mapsto \alpha_S(\mathbf{H}) \quad \text{and} \quad \mathbf{U} \mapsto \alpha_S(\mathbf{U})$$

do define a functor $\alpha_S: \widehat{\mathcal{C}}/\mathbf{h}_S \rightarrow \widehat{\mathcal{C}/S}$.

(ii) **Proposition 1.4.1.** *The functor $\alpha_S: \widehat{\mathcal{C}}/\mathbf{h}_S \rightarrow \widehat{\mathcal{C}/S}$ is an equivalence of categories.*

Let us merely indicate the principle of the construction of a quasi-inverse functor $\beta_S: \widehat{\mathcal{C}/S} \rightarrow \widehat{\mathcal{C}}/\mathbf{h}_S$. Let $\mathbf{G}$ be a functor on $\mathcal{C}/S$; for every object T of $\mathcal{C}$, one sets

$$\beta_S(\mathbf{G})(T) = \text{sum of the sets } \mathbf{G}(f) \text{ for } f \in \text{Hom}(T, S) = \mathbf{h}_S(T),$$

which defines a functor $\beta_S(\mathbf{G})$ on $\mathcal{C}$, equipped with an obvious projection onto $\mathbf{h}_S$.

1.5. — *The equivalence α_S commutes with the functors Γ .* In other words, if $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ is an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$ and $\alpha_S(\mathbf{H})$ the corresponding object of $\widehat{\mathcal{C}/S}$, one has

$$\Gamma(\alpha_S(\mathbf{H})) \simeq \Gamma(\mathbf{H}) \simeq \Gamma(\mathbf{F}/\mathbf{h}_S).$$

The equivalence α_S commutes with the functors $\mathbf{h}$, i.e. if $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, $\mathbf{h}_f: \mathbf{h}_T \rightarrow \mathbf{h}_S$ is an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$ whose transform by α_S is none other than $\mathbf{h}_{\mathcal{C}/S}(f)$, where

$$\mathbf{h}_{\mathcal{C}/S}: \mathcal{C}/S \rightarrow \widehat{\mathcal{C}/S}$$

is the canonical functor⁽¹¹⁾. Consequently:

⁽¹⁰⁾**N.D.E.** For example, if $\mathbf{F} = \mathbf{h}_X$, then $\mathbf{H}$ corresponds to a morphism $h: X \rightarrow S$ and $\mathbf{H}(T): \text{Hom}_{\mathcal{C}}(T, X) \rightarrow \text{Hom}_{\mathcal{C}}(T, S)$ is the map $g \mapsto h \circ g$, whence $\alpha_S(\mathbf{h}_X) = \text{Hom}_{\mathcal{C}/S}(-, X)$.

⁽¹¹⁾**N.D.E.** Cf. N.D.E. (10).

Proposition 1.5.1. *Let $\mathbf{H}: \mathbf{F} \rightarrow \mathbf{h}_S$ be an object of $\widehat{\mathcal{C}}/\mathbf{h}_S$. For $\alpha_S(\mathbf{H}): (\mathcal{C}/S)^\circ \rightarrow (\text{Ens})$ to be representable, it is necessary and sufficient that $\mathbf{F}: \mathcal{C}^\circ \rightarrow (\text{Ens})$ be representable; if $\mathbf{F} \simeq \mathbf{h}_T$, then $\alpha_S(\mathbf{H})$ is representable by the object $T \rightarrow S$ of $\mathcal{C}/S$.*

The equivalence α_S is transitive in S : if $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, one has a commutative diagram of equivalences

$$\begin{array}{ccccc} (\widehat{\mathcal{C}}/\mathbf{h}_S)/\mathbf{h}_T & \xrightarrow{\alpha_S/\mathbf{h}_T} & \widehat{(\mathcal{C}/S)}/\mathbf{h}_T & \xrightarrow{\alpha_f} & \widehat{(\mathcal{C}/S)}/T, \\ & \searrow \simeq & & & \swarrow \simeq \\ & & \widehat{\mathcal{C}}/\mathbf{h}_T & \xrightarrow{\alpha_T} & \widehat{\mathcal{C}}/T \end{array}$$

where $\alpha_S/\mathbf{h}_T$ denotes (provisionally) the *restriction* (cf. 1.6) of the functor α_S to the objects over $\mathbf{h}_T$.

1.6. Base change in a functor. — For every $S \in \text{Ob}(\mathcal{C})$, one has a canonical functor

$$i_S: \mathcal{C}/S \rightarrow \mathcal{C}$$

defined by $i_S(f) = T$ if f is the arrow $T \rightarrow S$. If $f: T \rightarrow S$ is an object of $\mathcal{C}/S$, one denotes $i_{T/S} = i_f$ the functor:

$$i_{T/S}: (\mathcal{C}/S)/T \rightarrow \mathcal{C}/S,$$

and one has the commutative diagram:

$$\begin{array}{ccccc} (\mathcal{C}/S)/T & \xrightarrow{i_{T/S}} & \mathcal{C}/S & \xrightarrow{i_S} & \mathcal{C} . \\ & \searrow \simeq & & & \swarrow i_T \\ & & \mathcal{C}/T & & \end{array}$$

That is, identifying $(\mathcal{C}/S)/T$ with $\mathcal{C}/T$ as we shall do henceforth,

$$i_S \circ i_{T/S} = i_T.$$

In the same way, if one identifies $\mathcal{C}$ and $\mathcal{C}/e$, when $\mathcal{C}$ possesses a final object e , then $i_{S/e}: \mathcal{C}/S \rightarrow \mathcal{C}/e$ is identified with i_S .

For $X \in \text{Ob}(\mathcal{C})$ (resp. $Y \in \text{Ob}(\mathcal{C}/S)$), let $p_S(X)$ (resp. $p_{T/S}(Y)$) be the object of $\mathcal{C}/S$ (resp. of $\mathcal{C}/T$) when it exists, defined by $X \times S$ (resp. $Y \times_S T$) equipped with its second projection:

$$\begin{array}{ccc} X \times S & & Y \times_S T \\ p_S(X) \downarrow & \text{resp.} & \downarrow p_{T/S}(Y) \\ S & & T \end{array}$$

The (partially defined) functor p_S (resp. $p_{T/S}$) is called the *base change functor*. It is, by definition of the product (resp. of the fibered product), the functor right adjoint to the functor i_S (resp. $i_{T/S}$)⁽¹²⁾. One likewise denotes

$$p_S(X) = X_S \quad \text{and} \quad p_{T/S}(Y) = Y_T.$$

The functor i_S defines a functor (*restriction*)

$$i_S^*: \widehat{\mathcal{C}} \rightarrow \widehat{\mathcal{C}}/S;$$

one denotes $\mathbf{F}_S = i_S^*(\mathbf{F}) = \mathbf{F} \circ i_S$. One evidently has

$$i_{T/S}^* \circ i_S^* = i_T^*,$$

⁽¹²⁾**N.D.E.** I.e., if $g: U \rightarrow S$ (resp. $h: V \rightarrow T$) is an object of $\mathcal{C}/S$ (resp. $\mathcal{C}/T$), then $i_S(g) = U$ and $i_{T/S}(h)$ is the object $f \circ h: V \rightarrow S$ of $\mathcal{C}/S$, and one has $\text{Hom}_{\mathcal{C}/S}(U, X \times S) \simeq \text{Hom}_{\mathcal{C}}(U, X)$, resp. $\text{Hom}_{\mathcal{C}/T}(V, X \times_S T) \simeq \text{Hom}_{\mathcal{C}/S}(V, X)$.

that is, for every functor $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$,

$$(\mathbf{F}_S)_T = \mathbf{F}_T.$$

The notation calls for a justification, which is as follows:

Proposition 1.6.1. *For the functor $(\mathbf{h}_X)_S: (\mathcal{C}/S)^\circ \longrightarrow (\text{Ens})$ to be representable, it is necessary and sufficient that the product $X \times S$ exist. One then has*

$$(\mathbf{h}_X)_S \simeq \mathbf{h}_{X_S}.$$

This shows that $\mathbf{F}_S$ has two interpretations: *restriction* of the functor $\mathbf{F}$ to $\mathcal{C}/S$, functor obtained by *base change* $\mathbf{e} \leftarrow S$. This leads to the following notation:

$$\begin{array}{ccccc} \mathbf{F} & & \mathbf{F}_S & & \mathbf{F}_T \\ & \swarrow & & \searrow & \\ \mathbf{e} & \longleftarrow & S & \longleftarrow & T \end{array}$$

which accounts well for the two preceding interpretations.

Let us observe that one has

$$\Gamma(\mathbf{F}_S) \simeq \text{Hom}(\mathbf{h}_S, \mathbf{F}) \simeq \mathbf{F}(S),$$

in particular

$$\Gamma(X_S) \simeq \text{Hom}(S, X).$$

1.7.0. — ⁽¹³⁾Let $\mathbf{E}$ be an object of $\widehat{\mathcal{C}}$. Let us consider the category $\mathcal{C}/\mathbf{E}$ of objects of $\mathcal{C}$ over $\mathbf{E}$: its objects are the pairs (V, ρ) consisting of an object V of $\mathcal{C}$ and of a $\widehat{\mathcal{C}}$ -morphism $\rho: \mathbf{h}_V \longrightarrow \mathbf{E}$, i.e. $\rho \in \mathbf{E}(V)$; a morphism from (V, ρ) to (V', ρ') is the datum of a $\mathcal{C}$ -morphism $f: V \longrightarrow V'$ such that $\rho' \circ f = \rho$ (i.e. $\mathbf{E}(f)(\rho') = \rho$). Let us denote by $\mathbf{L}$ the functor

$$\varinjlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{h}_V,$$

i.e. for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{L}(S) = \varinjlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{h}_V(S)$ is the set of equivalence classes of triples (V, ρ, v) , where $v: S \longrightarrow V$ is a $\mathcal{C}$ -morphism, and where one identifies (V, ρ, v) with $(V', \rho', f \circ v)$ for every $\mathcal{C}$ -morphism $f: V \longrightarrow V'$ such that $\rho' \circ f = \rho$.

Then the map $\varphi_{\mathbf{E}}(S)$ which, to the class of (V, ρ, v) , associates the element $\rho \circ v$ of $\mathbf{E}(S)$ is well-defined, and defines a morphism of functors

$$\varphi_{\mathbf{E}}: \varinjlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{h}_V \longrightarrow \mathbf{E}.$$

Lemma. $\varphi_{\mathbf{E}}$ is an isomorphism.

Indeed, let $S \in \text{Ob}(\mathcal{C})$. Every $x \in \mathbf{E}(S)$ is the image under $\varphi_{\mathbf{E}}(S)$ of the triple (S, x, id_S) ; this shows that $\varphi_{\mathbf{E}}(S)$ is surjective. On the other hand, let $\ell_1 = (V_1, \rho_1, v_1)$ and $\ell_2 = (V_2, \rho_2, v_2)$ be two elements of $\mathbf{L}(S)$ having the same image in $\mathbf{E}(S)$; set $\gamma = \rho_1 \circ v_1 = \rho_2 \circ v_2$. Then ℓ_1 and ℓ_2 are both equal, in $\mathbf{L}(S)$, to the class of the triple (S, γ, id_S) . This shows that $\varphi_{\mathbf{E}}(S)$ is injective.

Corollary. For every object $\mathbf{F}$ of $\widehat{\mathcal{C}}$, one has

$$\text{Hom}(\mathbf{E}, \mathbf{F}) = \varprojlim_{(V, \rho) \in \widehat{\mathcal{C}}_{\mathbf{E}}} \mathbf{F}(V).$$

⁽¹³⁾**N.D.E.** The lemma below has been added (cf. SGA 4, I.3.4); it is used in the proof of 1.7.1 and will be useful several times in the sequel.

1.7. — Hom, Isom, etc., objects. Let $\mathbf{F}$ and $\mathbf{G}$ be two objects of $\widehat{\mathcal{C}}$. We are going to define another object of $\widehat{\mathcal{C}}$ in the following way:

$$\begin{aligned}\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})(S) &= \text{Hom}_{\widehat{\mathcal{C}}/S}(\mathbf{F}_S, \mathbf{G}_S) \\ &\simeq \text{Hom}_{\widehat{\mathcal{C}}/\mathbf{h}_S}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G} \times \mathbf{h}_S) \simeq \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G}).\end{aligned}$$

The object $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ defined above possesses the following properties:

- (i) $\underline{\text{Hom}}(e, \mathbf{G}) \simeq \mathbf{G}$.⁽¹⁵⁾
- (ii) *The formation of $\underline{\text{Hom}}$ commutes with extension of the base:*

$$\underline{\text{Hom}}(\mathbf{F}_S, \mathbf{G}_S) \simeq \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})_S.$$

- (iii) $(\mathbf{F}, \mathbf{G}) \mapsto \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ is a bifunctor, contravariant in $\mathbf{F}$ and covariant in $\mathbf{G}$.

These three properties are obvious from the definitions.

We are going to show that, for every object $\mathbf{E}$ of $\widehat{\mathcal{C}}$, one has

$$\underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

Let $\varphi: \mathbf{E} \times \mathbf{F} \rightarrow \mathbf{G}$; we must associate with it a morphism from $\mathbf{E}$ to $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$. Let therefore $S' \rightarrow S$ be an arrow of $\mathcal{C}$. One has maps

$$\mathbf{E}(S) \times \mathbf{F}(S') \rightarrow \mathbf{E}(S') \times \mathbf{F}(S') \xrightarrow{\varphi(S')} \mathbf{G}(S').$$

Every element e of $\mathbf{E}(S)$ therefore defines, for every $S' \rightarrow S$, a map $\mathbf{F}(S') \rightarrow \mathbf{G}(S')$ functorial in S' , i.e. an element $\theta_\varphi(e)$ of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})(S)$. One has therefore obtained a map

$$\varphi \mapsto \theta_\varphi, \quad \text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \rightarrow \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})),$$

which is “functorial in $\mathbf{E}$ ”.

Proposition 1.7.1. ⁽¹⁶⁾ *Let $\mathbf{E}, \mathbf{F}, \mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$.*

- (a) *The map $\varphi \mapsto \theta_\varphi$ is a bijection:*

$$\text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \xrightarrow{\sim} \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})).$$

- (b) *Moreover, one has an isomorphism of functors:*

$$\underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

(a) Let us consider the two sides as functors in $\mathbf{E}$. The stated result is true if $\mathbf{E} = \mathbf{h}_X$; indeed, in this case it is nothing other than the definition of the functor $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$. On the other hand, the two sides, as functors in $\mathbf{E}$, transform inductive limits into projective limits. Finally, by lemma 1.7.0, every object $\mathbf{E}$ of $\widehat{\mathcal{C}}$ is isomorphic to the inductive limit of the $\mathbf{h}_X$, where X runs through the category $\mathcal{C}/\mathbf{E}$. This proves (a).

Let us sketch a direct proof of (a). To every $\theta \in \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G}))$, one associates the element φ_θ of $\text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G})$ defined as follows. For every $S \in \text{Ob}(\mathcal{C})$, one has a map

$$\theta(S): \mathbf{E}(S) \rightarrow \text{Hom}(\mathbf{F} \times S, \mathbf{G}),$$

functorial in S . If $(e, f) \in \mathbf{E}(S) \times \mathbf{F}(S)$, then f is a morphism $S \rightarrow \mathbf{F}$, hence $f \times \text{id}_S$ is a morphism $S \rightarrow \mathbf{F} \times S$; on the other hand, $\theta(S)(e)$ is a morphism $\mathbf{F} \times S \rightarrow \mathbf{G}$, hence by composition one obtains a morphism

$$\theta(S)(e) \circ (f \times \text{id}_S): S \rightarrow \mathbf{G},$$

i.e. an element $\varphi_\theta(S)(e, f)$ of $\mathbf{G}(S)$. One checks easily that the correspondence $S \mapsto \varphi_\theta(S)$ is functorial in S , hence defines a morphism φ_θ from $\mathbf{E} \times \mathbf{F}$ to $\mathbf{G}$. One leaves it to the reader to check that the maps $\theta \mapsto \varphi_\theta$ and $\varphi \mapsto \theta_\varphi$ are bijections inverse to one another.

⁽¹⁵⁾**N.D.E.** And, if $\mathbf{E}$ is a third object of $\widehat{\mathcal{C}}$, one has $\text{Hom}(\mathbf{E}, \mathbf{F} \times \mathbf{G}) \simeq \text{Hom}(\mathbf{E}, \mathbf{F}) \times \text{Hom}(\mathbf{E}, \mathbf{G})$.

⁽¹⁶⁾**N.D.E.** One has added point (b), which will be useful in II.1 and II.3.11. On the other hand, one has sketched a second, more direct, proof of point (a).

Let us prove (b). If $S \in \text{Ob}(\mathcal{C})$, one has, by 1.7 (ii) and (a) applied to $\mathcal{C}/S$:

$$\begin{aligned} \underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G}))(S) &\simeq \text{Hom}_S(\mathbf{E}_S, \underline{\text{Hom}}_S(\mathbf{F}_S, \mathbf{G}_S)) \\ &\simeq \text{Hom}_S(\mathbf{E}_S \times_S \mathbf{F}_S, \mathbf{G}_S) \\ &\simeq \text{Hom}(\mathbf{E} \times \mathbf{F} \times S, \mathbf{G}) \\ &\simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G})(S) \end{aligned}$$

and these isomorphisms are functorial in S .

Corollary 1.7.2. *One has:*

$$\begin{aligned} \underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) &\simeq \underline{\text{Hom}}(\mathbf{F}, \underline{\text{Hom}}(\mathbf{E}, \mathbf{G})), \\ \underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) &\simeq \underline{\text{Hom}}(\mathbf{F}, \underline{\text{Hom}}(\mathbf{E}, \mathbf{G})). \end{aligned}$$

In particular, setting $\mathbf{E} = e$, and taking into account $\underline{\text{Hom}}(e, \mathbf{G}) \simeq \mathbf{G}$, one has

$$\Gamma(\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \text{Hom}(\mathbf{F}, \mathbf{G}).$$

Let us note that the composition of the $\underline{\text{Hom}}$ furnishes functorial morphisms

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{H}) \longrightarrow \underline{\text{Hom}}(\mathbf{F}, \mathbf{H}).$$

If $\mathbf{F}$ and $\mathbf{G}$ are two objects of $\widehat{\mathcal{C}}$, one denotes by $\text{Isom}(\mathbf{F}, \mathbf{G})$ the subset of $\text{Hom}(\mathbf{F}, \mathbf{G})$ consisting of the isomorphisms from $\mathbf{F}$ onto $\mathbf{G}$. One then defines a subobject $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$ of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ by:

$$\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})(S) = \text{Isom}(\mathbf{F}_S, \mathbf{G}_S).$$

One then has isomorphisms

$$\begin{aligned} \Gamma(\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})) &\simeq \text{Isom}(\mathbf{F}, \mathbf{G}), \\ \underline{\text{Isom}}(\mathbf{F}, \mathbf{G}) &\simeq \underline{\text{Isom}}(\mathbf{G}, \mathbf{F}). \end{aligned}$$

In the particular case where $\mathbf{F} = \mathbf{G}$, one sets

$$\begin{aligned} \underline{\text{End}}(\mathbf{F}) &= \underline{\text{Hom}}(\mathbf{F}, \mathbf{F}), \quad \text{End}(\mathbf{F}) = \text{Hom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\text{End}}(\mathbf{F})), \\ \underline{\text{Aut}}(\mathbf{F}) &= \underline{\text{Isom}}(\mathbf{F}, \mathbf{F}), \quad \text{Aut}(\mathbf{F}) = \text{Isom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\text{Aut}}(\mathbf{F})). \end{aligned}$$

The formation of the objects $\underline{\text{Hom}}$, $\underline{\text{Isom}}$, $\underline{\text{Aut}}$, $\underline{\text{End}}$ commutes with base changes.

Remark 1.7.3. ⁽¹⁷⁾ Let us observe that one can construct an object isomorphic to $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$ in the following way: one has a morphism

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\text{End}}(\mathbf{F});$$

interchanging $\mathbf{F}$ and $\mathbf{G}$, one deduces from it a morphism

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G}).$$

On the other hand, the identity morphism of $\mathbf{F}$ is an element of $\underline{\text{End}}(\mathbf{F})$ and therefore defines a morphism $e \longrightarrow \underline{\text{End}}(\mathbf{F})$. Doing the same with $\mathbf{G}$ and taking the product, one finds a morphism

$$e \longrightarrow \underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G}).$$

It is then immediate that the fiber product of e and of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F})$ over $\underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G})$ is isomorphic to $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$.

All these definitions apply in particular to the case where $\mathbf{F} = \mathbf{h}_X$, $\mathbf{G} = \mathbf{h}_Y$. In the case where $\underline{\text{Hom}}(\mathbf{h}_X, \mathbf{h}_Y)$ is representable by an object of $\mathcal{C}$, one denotes this object by $\text{Hom}(X, Y)$. It possesses the following property: if $Z \times X$ exists, then

$$\text{Hom}(Z, \text{Hom}(X, Y)) \simeq \text{Hom}(Z \times X, Y).$$

This property characterizes it when products exist in $\mathcal{C}$.

⁽¹⁷⁾**N.D.E.** One has added the numbering 1.7.3, for later references.

One likewise defines (whenever they are willing to exist) objects

$$\text{Isom}(X, Y), \quad \text{End}(X), \quad \text{Aut}(X);$$

let us simply observe that, by the construction given above, $\text{Isom}(X, Y)$ exists whenever fiber products exist in $\mathcal{C}$ and $\text{Hom}(X, Y)$, $\text{Hom}(Y, X)$, $\text{End}(X)$ and $\text{End}(Y)$ exist.

Everything that precedes applies equally to categories of the form $\mathcal{C}/S$. The corresponding objects will be denoted as explicitly as possible by appropriate symbols: for example, if T and T' are two objects of $\mathcal{C}$ over S , one will denote by $\text{Hom}_S(T, T')$ the object $\text{Hom}_{\mathcal{C}/S}(T/S, T'/S)$.

1.8. — Constant objects. Let $\mathcal{C}$ be a category in which direct sums and fiber products exist and in which direct sums commute with base changes (for example the category of schemes⁽¹⁸⁾). For every set E and for every object S of $\mathcal{C}$, let us set

$$(1.8.1) \quad E_S = \begin{cases} \text{the direct sum of a family } (S_i)_{i \in E} \\ \text{of objects of } \mathcal{C} \text{ all isomorphic to } S. \end{cases}$$

This object is characterized by the formula:

$$(1.8.2) \quad \text{Hom}_{\mathcal{C}}(E_S, T) = \text{Hom}(E, \text{Hom}_{\mathcal{C}}(S, T)),$$

for every $T \in \text{Ob}(\mathcal{C})$, where the second Hom is taken in the category of sets.

The object E_S is equipped with a canonical projection onto S , in such a way that $E \mapsto E_S$ is in fact a functor from (Ens) to $\mathcal{C}/S$.

⁽¹⁹⁾ If $S' \rightarrow S$ is an arrow of $\mathcal{C}$, one has, since direct sums commute with base changes,

$$E_{S'} = (E_S)_{S'}.$$

In particular, if $\mathcal{C}$ possesses a final object e , one has

$$E_S = (E_e)_S.$$

The functor $E \mapsto E_S/S$, from (Ens) to $\mathcal{C}/S$, commutes with finite products. It suffices, for this, to see that

$$(\times) \quad E_S \times_S F_S = (E \times F)_S.$$

Now, by the results of 1.7 applied to $\mathcal{C}/S$, one has, for every $T \in \text{Ob}(\mathcal{C}/S)$, natural isomorphisms (all unspecified Hom being taken in $\mathcal{C}/S$):

$$\begin{aligned} \text{Hom}((E \times F)_S, T) &\simeq \text{Hom}_{(\text{Ens})}(E \times F, \text{Hom}(S, T)) \\ &\simeq \text{Hom}_{(\text{Ens})}(E, \text{Hom}_{(\text{Ens})}(F, \text{Hom}(S, T))) \simeq \text{Hom}_{(\text{Ens})}(E, \text{Hom}(F_S, T)) \end{aligned}$$

and

$$\begin{aligned} \text{Hom}(E_S \times_S F_S, T) &\simeq \text{Hom}(E_S, \text{Hom}(F_S, T)) \\ &\simeq \text{Hom}_{(\text{Ens})}(E, \text{Hom}(S, \text{Hom}(F_S, T))). \end{aligned}$$

Now, $\text{Hom}(S, \text{Hom}(F_S, T)) = \text{Hom}(F_S, T)$, whence $(\times)$.

Suppose that, $\emptyset$ denoting an initial object of $\mathcal{C}$, the diagram

$$(*) \quad \begin{array}{ccc} \emptyset & \longrightarrow & S \\ \downarrow & & \downarrow \\ S & \longrightarrow & S \amalg S \end{array}$$

is cartesian. (This is the case for the category of schemes.)⁽²⁰⁾ Then the functor $E \mapsto E_S$ commutes with finite projective limits.

⁽¹⁸⁾**N.D.E.** One has replaced the old terminology “preschemes/schemes” by the current terminology “schemes/separated schemes”.

⁽¹⁹⁾**N.D.E.** In the three paragraphs which follow, one has modified the order of the sentences and added a few clarifications concerning the role of hypothesis $(*)$ below.

⁽²⁰⁾**N.D.E.** $(*)$ is *not* satisfied if $\mathcal{C}$ is the category whose arrows are $A \rightarrow B$ and id_A, id_B ; in this case $B \amalg B = B$ and $B \times_B B = B \neq A$.

Indeed, taking $(\times)$ into account, it suffices to see that $E \mapsto E_S$ commutes with fiber products. Let $u: E \rightarrow G$ and $v: F \rightarrow G$ be two maps of sets. Since in $\mathcal{C}$ direct sums commute with base changes, one has

$$(1) \quad F_S \times_{G_S} E_S \simeq \coprod_{f \in F} S_f \times_{G_S} E_S \simeq \coprod_{\substack{f \in F \\ x \in E}} S_f \times_{G_S} S_x.$$

If $v(f) \neq u(x)$, there exists in $\mathcal{C}/S$ a morphism

$$S_f \times_{G_S} S_x \longrightarrow S_f \times_{S_{v(f)} \coprod S_{u(x)}} S_x;$$

now, by hypothesis $(*)$, the right-hand term is $\emptyset$. Consequently,

$$(2) \quad S_f \times_{G_S} S_x \simeq \emptyset \quad \text{if } v(f) \neq u(x).$$

On the other hand, if $v(f) = u(x)$, there exists in $\mathcal{C}/S$ a morphism

$$S \longrightarrow S_f \times_{G_S} S_x;$$

since $S \xrightarrow{\text{id}} S$ is a final object of $\mathcal{C}/S$, it follows that

$$(3) \quad S_f \times_{G_S} S_x \simeq S \quad \text{if } v(f) = u(x).$$

Combining (1), (2) and (3), one obtains an isomorphism functorial in S

$$F_S \times_{G_S} E_S \simeq \coprod_{F \times_G E} S = (F \times_G E)_S.$$

An object of the form E_S will be called a *constant object*. Let us observe that one has a morphism functorial in E :

$$E \longrightarrow \Gamma(E_S/S)$$

which associates with each $i \in E$ the section of E_S over S defined by the isomorphism of S onto S_i . Suppose condition $(*)$ satisfied for every object S of $\mathcal{C}$; then the morphism $E \rightarrow \Gamma(E_S/S)$ is a monomorphism for every $S \neq \emptyset$.

If $\mathcal{C}$ is the category of schemes, then $\Gamma(E_S/S)$ is identified with the locally constant maps from the topological space S to the set E , the preceding map associating with each element of E the corresponding constant map. Let us observe that it follows from what one has just said that E_S may also be defined as representing the functor which, to every S' over S , associates the set of locally constant functions from the topological space S' to the set E .⁽²¹⁾

2. Algebraic structures

Given a species of algebraic structure in the category of sets, we propose to extend it to the category $\mathcal{C}$. Let us first treat an example: the case of groups.

2.1. — Group structures. We keep the notations of the preceding paragraph.

Definition 2.1.1. Let $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$. One calls structure of $\widehat{\mathcal{C}}$ -group on $\mathbf{G}$ the datum, for every $S \in \text{Ob}(\mathcal{C})$, of a group structure on the set $\mathbf{G}(S)$, in such a way that for every arrow $f: S' \rightarrow S''$ of $\mathcal{C}$, the map $\mathbf{G}(f): \mathbf{G}(S'') \rightarrow \mathbf{G}(S')$ is a group homomorphism. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, one calls morphism of $\widehat{\mathcal{C}}$ -groups from $\mathbf{G}$ to $\mathbf{H}$ every morphism $u \in \text{Hom}(\mathbf{G}, \mathbf{H})$ such that for every $S \in \text{Ob}(\mathcal{C})$ the map of sets $u(S): \mathbf{G}(S) \rightarrow \mathbf{H}(S)$ is a group homomorphism.

One denotes by $\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H})$ the set of morphisms of $\widehat{\mathcal{C}}$ -groups from $\mathbf{G}$ to $\mathbf{H}$ and by $(\widehat{\mathcal{C}}\text{-Gr.})$ the category of $\widehat{\mathcal{C}}$ -groups.

⁽²¹⁾**N.D.E.** Let us also note that the diagonal morphism $E_S \rightarrow E_S \times_S E_S = (E \times E)_S$ is a closed immersion, i.e. E_S is separated over S .

Examples. Let $\mathbf{E} \in \text{Ob}(\widehat{\mathcal{C}})$; the object $\underline{\text{Aut}}(\mathbf{E})$ is equipped in an evident way with a structure of $\widehat{\mathcal{C}}$ -group. The final object e possesses a unique structure of $\widehat{\mathcal{C}}$ -group which makes it a final object of $(\widehat{\mathcal{C}}\text{-Gr.})$.

For every $S \in \text{Ob}(\mathcal{C})$, let $e_{\mathbf{G}}(S)$ be the unit element of $\mathbf{G}(S)$. The family of the $e_{\mathbf{G}}(S)$ defines an element $e_{\mathbf{G}} \in \Gamma(\mathbf{G}) = \text{Hom}(e, \mathbf{G})$ which is a morphism of $\widehat{\mathcal{C}}$ -groups $e \rightarrow \mathbf{G}$ and which one calls the *unit section* of $\mathbf{G}$. Let us observe that giving a structure of $\widehat{\mathcal{C}}$ -group on $\mathbf{G}$ amounts to giving a law of composition on $\mathbf{G}$, that is, a $\widehat{\mathcal{C}}$ -morphism

$$\pi_{\mathbf{G}}: \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$$

such that for every $S \in \text{Ob}(\mathcal{C})$, $\pi_{\mathbf{G}}(S)$ equips $\mathbf{G}(S)$ with a group structure.

In the same way, $f: \mathbf{G} \rightarrow \mathbf{H}$ is a morphism of $\widehat{\mathcal{C}}$ -groups if and only if the following diagram is commutative:

$$\begin{array}{ccc} \mathbf{G} \times \mathbf{G} & \xrightarrow{\pi_{\mathbf{G}}} & \mathbf{G} \\ (f, f) \downarrow & & \downarrow f \\ \mathbf{H} \times \mathbf{H} & \xrightarrow{\pi_{\mathbf{H}}} & \mathbf{H}. \end{array}$$

A subobject $\mathbf{H}$ of $\mathbf{G}$ such that, for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{H}(S)$ is a subgroup of $\mathbf{G}(S)$ evidently possesses a structure of $\widehat{\mathcal{C}}$ -group induced by that of $\mathbf{G}$: it is the only one for which the monomorphism $\mathbf{H} \rightarrow \mathbf{G}$ is a morphism of $\widehat{\mathcal{C}}$ -groups. The $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ equipped with this structure is called a sub- $\widehat{\mathcal{C}}$ -group of $\mathbf{G}$.

If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, the product $\mathbf{G} \times \mathbf{H}$ is equipped with an evident structure of $\widehat{\mathcal{C}}$ -group: for every $S \in \text{Ob}(\mathcal{C})$, one equips $\mathbf{G}(S) \times \mathbf{H}(S)$ with the product group structure of the group structures given on $\mathbf{G}(S)$ and $\mathbf{H}(S)$. The $\widehat{\mathcal{C}}$ -group $\mathbf{G} \times \mathbf{H}$ equipped with this structure will be called the product $\widehat{\mathcal{C}}$ -group of $\mathbf{G}$ and of $\mathbf{H}$ (it is, moreover, the product in the category of $\widehat{\mathcal{C}}$ -groups).

If $\mathbf{G}$ is a $\widehat{\mathcal{C}}$ -group, then for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{G}_S$ is a $\widehat{\mathcal{C}}_S$ -group. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, one will define the object $\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H})$ of $\widehat{\mathcal{C}}$ by:

$$\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H})(S) = \text{Hom}_{\widehat{\mathcal{C}}_S\text{-Gr.}}(\mathbf{G}_S, \mathbf{H}_S).$$

(Note: $\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr.}}$ is not in general a $\widehat{\mathcal{C}}$ -group, nor a fortiori the object Hom in the category $(\widehat{\mathcal{C}}\text{-Gr.})$.)

One defines in the same way the objects

$$\text{Isom}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}, \mathbf{H}), \quad \text{End}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}), \quad \text{Aut}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}).$$

Definition 2.1.2. Let $G \in \text{Ob}(\mathcal{C})$. One calls a structure of $\widehat{\mathcal{C}}$ -group on $\mathbf{h}_G \in \text{Ob}(\widehat{\mathcal{C}})$ a structure of $\mathcal{C}$ -group on G . One calls an element $u \in \text{Hom}(G, H) \simeq \text{Hom}(\mathbf{h}_G, \mathbf{h}_H)$ which defines a morphism of $\widehat{\mathcal{C}}$ -group from $\mathbf{h}_G$ to $\mathbf{h}_H$ a morphism from the $\mathcal{C}$ -group G to the $\mathcal{C}$ -group H .

One denotes by $(\mathcal{C}\text{-Gr.})$ the category of $\mathcal{C}$ -groups. Let us note that there exists in (Cat) a cartesian square

$$\begin{array}{ccc} (\mathcal{C}\text{-Gr.}) & \longrightarrow & (\widehat{\mathcal{C}}\text{-Gr.}) \\ \downarrow & & \downarrow \\ \mathcal{C} & \xrightarrow{\mathbf{h}} & \widehat{\mathcal{C}}. \end{array}$$

All the preceding definitions and constructions are therefore transported at once to $(\mathcal{C}\text{-Gr.})$ whenever the functors which they involve (products, Hom objects, etc.) are representable. They also apply to the categories $\mathcal{C}_S$. In this case, we shall write $\text{Hom}_{S\text{-Gr.}}$ for $\text{Hom}_{\widehat{\mathcal{C}}_S\text{-Gr.}}$, etc.

2.2. — More generally, if (T) is a species of structure on n base sets defined by finite projective limits (for example, by commutativities of diagrams constructed with cartesian products: structures of monoid, of group, of set with operators, of module over a ring, of Lie algebra over a ring, etc.), the preceding construction makes it possible to define the notion of “structure of species (T) on n objects $\mathbf{F}_1, \dots, \mathbf{F}_n$ of $\widehat{\mathcal{C}}$ ”: such a structure will be the giving, for each S of $\mathcal{C}$, of a structure of species (T) on the sets $\mathbf{F}_1(S), \dots, \mathbf{F}_n(S)$ in such a way that for every arrow $S' \rightarrow S''$ of $\mathcal{C}$, the family of maps $(\mathbf{F}_i(S'')) \rightarrow (\mathbf{F}_i(S'))$ is a poly-homomorphism for the species of structure (T) . One defines in a similar manner the morphisms of the species of structure (T) , whence a category $(\widehat{\mathcal{C}} \times \widehat{\mathcal{C}} \times \dots \times \widehat{\mathcal{C}})^{(T)}$. The fully faithful functor $(\mathbf{h} \times \mathbf{h} \times \dots \times \mathbf{h})$ then makes it possible to define by inverse image the category $(\mathcal{C} \times \mathcal{C} \times \dots \times \mathcal{C})^{(T)}$, then, as it commutes with projective limits, to transport thereto all the functorial properties, notions and notations introduced in $\widehat{\mathcal{C}}$. Suppose now that fibered products exist in $\mathcal{C}$, and let (T) be a species of algebraic structure defined by the *giving* of certain morphisms between cartesian products satisfying *axioms* consisting of certain commutativities of diagrams constructed by means of the preceding arrows. A structure of species (T) on a family of objects of $\mathcal{C}$ will therefore be defined by certain morphisms between cartesian products satisfying certain commutation conditions. It follows that if $\mathcal{C}$ and $\mathcal{C}'$ are two categories possessing products and $f: \mathcal{C} \rightarrow \mathcal{C}'$ is a functor *commuting with products*, then for every family of objects (F_i) of $\mathcal{C}$ equipped with a structure of species (T) , the family $(f(F_i))$ of objects of $\mathcal{C}'$ will thereby be equipped also with a structure of species (T) . Every $\mathcal{C}$ -group will be transformed into a $\mathcal{C}'$ -group, every pair $(\mathcal{C}$ -ring, $\mathcal{C}$ -module over this $\mathcal{C}$ -ring) into an analogous pair in $\mathcal{C}'$, etc.

Let in particular $\mathcal{C}$ be a category satisfying the conditions of 1.8⁽²¹⁾; the functor $E \mapsto E_S$ defined in *loc. cit.* commutes with finite projective limits; it therefore transforms group into S -group (i.e. $\mathcal{C}/S$ -group), ring into S -ring, etc.

Remark. It is well to remark that the preceding construction procedure, applied to the category $\widehat{\mathcal{C}}$, does indeed give back the notions which one has already defined there; in other words, it amounts to the same to give, on an object of $\widehat{\mathcal{C}}$, a structure of species (T) when one considers this object as a functor on $\mathcal{C}$, or to give a structure of species (T) on the representable functor on $\widehat{\mathcal{C}}$ defined by this object.⁽²²⁾

We shall now treat two further particular cases of the preceding construction, the case of structures with groups of operators and the case of modules.

2.3. Structures with groups of operators. —

Definition 2.3.1. Let $\mathbf{E} \in \text{Ob}(\widehat{\mathcal{C}})$ and $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}}\text{-Gr.})$. A structure of object with $\widehat{\mathcal{C}}$ -group of operators $\mathbf{G}$ (or of $\mathbf{G}$ -object) on $\mathbf{E}$ is the giving, on $\mathbf{E}(S)$, for every $S \in \text{Ob}(\mathcal{C})$, of a structure of set with group of operators $\mathbf{G}(S)$ in such a way that, for every arrow $S' \rightarrow S''$ of $\mathcal{C}$, the map of sets $\mathbf{E}(S'') \rightarrow \mathbf{E}(S')$ is compatible with the homomorphism of operators $\mathbf{G}(S'') \rightarrow \mathbf{G}(S')$.

As usual, it amounts to the same to give a morphism

$$\mu: \mathbf{G} \times \mathbf{E} \rightarrow \mathbf{E}$$

which for every S equips $\mathbf{E}(S)$ with a structure of set with operators $\mathbf{G}(S)$. But $\text{Hom}(\mathbf{G} \times \mathbf{E}, \mathbf{E}) \simeq \text{Hom}(\mathbf{G}, \text{End}(\mathbf{E}))$, hence μ defines a morphism $\mathbf{G} \rightarrow \text{End}(\mathbf{E})$ and it is immediate that the latter maps $\mathbf{G}$ into $\text{Aut}(\mathbf{E})$ and that it is a morphism of $\widehat{\mathcal{C}}$ -groups. Consequently: *to give on $\mathbf{E}$ a structure of object with $\widehat{\mathcal{C}}$ -group of operators $\mathbf{G}$ is equivalent to giving a morphism*

⁽²¹⁾N.D.E. Including the condition $(*)$.

⁽²²⁾N.D.E. For example, for the group structures: let $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$; if the functor $\widehat{\mathcal{C}} \rightarrow (\text{Ens})$, $\mathbf{F} \mapsto \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F}, \mathbf{G})$ is equipped with a group structure, the same is true of its restriction to $\mathcal{C}$, $X \mapsto \mathbf{G}(X)$. Conversely, if $\mathbf{G}$ is a $\widehat{\mathcal{C}}$ -group, then the “multiplication” morphism $\pi_{\mathbf{G}}: \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$ induces, for every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$, a group structure on $\text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F}, \mathbf{G})$, functorial in $\mathbf{F}$.

of $\widehat{\mathcal{C}}$ -groups

$$\rho: \mathbf{G} \longrightarrow \text{Aut}(\mathbf{E}).$$

In particular, every element $g \in \mathbf{G}(S)$ defines an automorphism $\rho(g)$ of the functor $\mathbf{E}_S$, that is, an automorphism of $\mathbf{E} \times \mathbf{h}_S$ commuting with the projection $\mathbf{E} \times \mathbf{h}_S \longrightarrow \mathbf{h}_S$, and in particular an automorphism of the set $\mathbf{E}(S')$ for every $S' \longrightarrow S$.

Definition 2.3.2. One denotes by $\mathbf{E}^{\mathbf{G}}$ the subobject of $\mathbf{E}$ defined as follows:

$$\mathbf{E}^{\mathbf{G}}(S) = \{x \in \mathbf{E}(S) \mid x_{S'} \text{ invariant under } \mathbf{G}(S') \text{ for every } S' \longrightarrow S\},$$

where $x_{S'}$ denotes the image of x by $\mathbf{E}(S) \longrightarrow \mathbf{E}(S')$.

Then $\mathbf{E}^{\mathbf{G}}$ (“subobject of the invariants of $\mathbf{G}$ ”) is the largest subobject of $\mathbf{E}$ on which $\mathbf{G}$ operates trivially.

Definition 2.3.3. Let $\mathbf{F}$ be a subobject of $\mathbf{E}$. One denotes by $\text{Norm}_{\mathbf{G}} \mathbf{F}$ and $\text{Centr}_{\mathbf{G}} \mathbf{F}$ the sub- $\widehat{\mathcal{C}}$ -groups of $\mathbf{G}$ defined by

$$\begin{aligned} (\text{Norm}_{\mathbf{G}} \mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}_S = \mathbf{F}_S\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}(S') = \mathbf{F}(S'), \text{ for every } S' \longrightarrow S\}, \\ (\text{Centr}_{\mathbf{G}} \mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}_S} = \text{identity}\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}(S')} = \text{identity}, \text{ for every } S' \longrightarrow S\}, \end{aligned}$$

where the vertical bar after $\rho(g)$ denotes restriction.⁽²³⁾

Scholium 2.3.3.1. ⁽²⁴⁾ In particular, let $x \in \Gamma(\mathbf{E})$, i.e. (cf. 1.2) a collection of elements $x_S \in \mathbf{E}(S)$, $S \in \text{Ob}(\mathcal{C})$, such that for every arrow $f: S' \longrightarrow S$ one has $\mathbf{E}(f)(x_S) = x_{S'}$ (if $\mathcal{C}$ possesses a final object S_0 one has $\Gamma(\mathbf{E}) = \mathbf{E}(S_0)$). Then x defines a subfunctor of $\mathbf{E}$, which one will denote by x , and one has $\text{Norm}_{\mathbf{G}} x = \text{Centr}_{\mathbf{G}} x$. One will denote by $\text{Stab}_{\mathbf{G}}(x)$, and one will call stabilizer of x , this functor; for every $S \in \text{Ob}(\mathcal{C})$ one therefore has:

$$\text{Stab}_{\mathbf{G}}(x)(S) = \{g \in \mathbf{G}(S) \mid \rho(g)x_S = x_S\}.$$

Suppose that fibered products exist in $\mathcal{C}$; if $\mathbf{G} = \mathbf{h}_G$ (resp. $\mathbf{E} = \mathbf{h}_E$), where G is a $\mathcal{C}$ -group (resp. $E \in \text{Ob}(\mathcal{C})$), and if $\mathcal{C}$ possesses a final object S_0 , so that x is a morphism $S_0 \longrightarrow E$, then $\text{Stab}_{\mathbf{G}}(x)$ is representable by the fibered product $G \times_E S_0$, where $G \longrightarrow E$ is the composite of $\text{id}_G \times x: G = G \times S_0 \longrightarrow G \times E$ and of $\mu: G \times E \longrightarrow E$.

Remark 2.3.3.2. ⁽²⁴⁾ The formation of $\mathbf{E}^{\mathbf{G}}$, $\text{Norm}_{\mathbf{G}} \mathbf{F}$ and $\text{Centr}_{\mathbf{G}} \mathbf{F}$ commutes with base change, i.e. for every $S \in \text{Ob}(\mathcal{C})$ one has

$$(\mathbf{E}^{\mathbf{G}})_S = (\mathbf{E}_S)^{\mathbf{G}_S}, \quad (\text{Norm}_{\mathbf{G}} \mathbf{F})_S \simeq \text{Norm}_{\mathbf{G}_S} \mathbf{F}_S, \quad (\text{Centr}_{\mathbf{G}} \mathbf{F})_S \simeq \text{Centr}_{\mathbf{G}_S} \mathbf{F}_S.$$

Definition 2.3.4. If G is a $\mathcal{C}$ -group and $\mathbf{E}$ an object of $\widehat{\mathcal{C}}$ (resp. E an object of $\mathcal{C}$), a structure of G -object on $\mathbf{E}$ (resp. on E) is a structure of $\mathbf{h}_G$ -object on $\mathbf{E}$ (resp. $\mathbf{h}_E$).

In view of this definition, all the notions and notations defined above carry over to $\mathcal{C}$, when they involve only representable functors: for example if $\text{Norm}_{\mathbf{h}_G}(\mathbf{h}_F)$ is representable, then there exists one and only one subobject of G which represents it and which is then a sub- $\mathcal{C}$ -group of G ; one denotes it by $\text{Norm}_G(F)$, etc.

Definition 2.3.5. a) One says that the $\widehat{\mathcal{C}}$ -group $\mathbf{G}$ operates on the $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ if $\mathbf{H}$ is equipped with a structure of $\mathbf{G}$ -object such that, for every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{H}(S)$ defined by g is a group automorphism.

⁽²³⁾**N.D.E.** One has corrected the original, replacing the inclusion $\rho(g)\mathbf{F}_S \subset \mathbf{F}_S$ by an equality, in order to ensure that $\text{Norm}_{\mathbf{G}}(\mathbf{F})$ is indeed a group (see VI_B 6.4 for conditions under which the “transporter” coincides with the “strict transporter”).

⁽²⁴⁾**N.D.E.** One has added scholium 2.3.3.1 and remark 2.3.3.2.

It amounts to the same to say that for every $g \in \mathbf{G}(S)$, the automorphism $\rho(g)$ of $\mathbf{H}_S$ is an automorphism of $\widehat{\mathcal{C}}/S$ -groups, or again that the morphism of $\widehat{\mathcal{C}}$ -groups $\mathbf{G} \rightarrow \text{Aut}(\mathbf{H})$ maps $\mathbf{G}$ into $\text{Aut}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{H})$.

b) In the above situation, there exists on the product $\mathbf{H} \times \mathbf{G}$ a unique structure of $\widehat{\mathcal{C}}$ -group such that, for every S , $(\mathbf{H} \times \mathbf{G})(S)$ is the semi-direct product of the groups $\mathbf{H}(S)$ and $\mathbf{G}(S)$ relative to the given operation of $\mathbf{G}(S)$ on $\mathbf{H}(S)$. One will denote this $\widehat{\mathcal{C}}$ -group by $\mathbf{H} \cdot \mathbf{G}$ and call it the *semi-direct product of $\mathbf{H}$ by $\mathbf{G}$* . One therefore has by definition

$$(\mathbf{H} \cdot \mathbf{G})(S) = \mathbf{H}(S) \cdot \mathbf{G}(S).$$

Let $\mathbf{G}$ be a $\widehat{\mathcal{C}}$ -group. For every arrow $S' \rightarrow S$ of $\mathcal{C}$ and every $g \in \mathbf{G}(S)$, let $\text{Int}(g)$ be the automorphism of $\mathbf{G}(S')$ defined by $\text{Int}(g)h = ghg^{-1}$. This definition extends to that of a morphism of $\widehat{\mathcal{C}}$ -groups

$$\text{Int}: \mathbf{G} \rightarrow \text{Aut}_{\widehat{\mathcal{C}}\text{-Gr.}}(\mathbf{G}) \subset \text{Aut}(\mathbf{G}).$$

Definition 2.3.3 therefore applies, and one has sub- $\widehat{\mathcal{C}}$ -groups of $\mathbf{G}$

$$\text{Norm}_{\mathbf{G}}(\mathbf{E}) \quad \text{and} \quad \text{Centr}_{\mathbf{G}}(\mathbf{E})$$

for every subobject $\mathbf{E}$ of $\mathbf{G}$.

Definition 2.3.6. One calls *center* of $\mathbf{G}$, and one denotes by $\text{Centr}(\mathbf{G})$, the sub- $\widehat{\mathcal{C}}$ -group $\text{Centr}_{\mathbf{G}}(\mathbf{G})$ of $\mathbf{G}$. One says that $\mathbf{G}$ is commutative if $\text{Centr}(\mathbf{G}) = \mathbf{G}$ or, what amounts to the same, if $\mathbf{G}(S)$ is commutative for every S .

One says that the sub- $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ of $\mathbf{G}$ is *invariant* in $\mathbf{G}$ if $\text{Norm}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, what amounts to the same, if $\mathbf{H}(S)$ is invariant in $\mathbf{G}(S)$ for every S .⁽²⁵⁾

Definition 2.3.6.1.⁽²⁶⁾ Let $f: \mathbf{G} \rightarrow \mathbf{G}'$ be a morphism of $\widehat{\mathcal{C}}$ -groups. One calls *kernel* of f , and one denotes by $\text{Ker } f$, the sub- $\widehat{\mathcal{C}}$ -group of $\mathbf{G}$ defined by

$$(\text{Ker } f)(S) = \{x \in \mathbf{G}(S) \mid f(S)(x) = 1\} = \text{Ker } f(S)$$

for every $S \in \text{Ob}(\mathcal{C})$; it is an *invariant* sub- $\widehat{\mathcal{C}}$ -group.

Let us note that if $\mathbf{G} = \mathbf{h}_G$ and $\mathbf{G}' = \mathbf{h}_{G'}$, if $\mathcal{C}$ possesses a final object S_0 and if fibered products exist in $\mathcal{C}$, then $\text{Ker}(f)$ is *representable* by $S_0 \times_{G'} G$.

Definition 2.3.6.2.⁽²⁶⁾ Let $\mathbf{E} \in \widehat{\mathcal{C}}$ and let $\mathbf{G}$ be a $\widehat{\mathcal{C}}$ -group operating on $\mathbf{E}$. One says that the operation of $\mathbf{G}$ on $\mathbf{E}$ is *faithful* if the kernel of the morphism $\mathbf{G} \rightarrow \text{Aut}(\mathbf{E})$ is trivial, i.e. if for every $S \in \text{Ob}(\mathcal{C})$ and $g \in \mathbf{G}(S)$, the condition $g_{S'} \cdot x = x$ for every $S' \rightarrow S$ and $x \in \mathbf{E}(S')$, implies $g = 1$.

Many definitions and propositions of the elementary theory of groups transpose easily. Let us simply point out the following one, which will be useful to us:

Proposition 2.3.7. Let $f: \mathbf{W} \rightarrow \mathbf{G}$ be a morphism of $\widehat{\mathcal{C}}$ -groups. Set $\mathbf{H}(S) = \text{Ker } f(S)$. Let $u: \mathbf{G} \rightarrow \mathbf{W}$ be a morphism of $\widehat{\mathcal{C}}$ -groups which is a section of f (and which is then necessarily a monomorphism). Then $\mathbf{W}$ is identified with the semi-direct product of $\mathbf{H}$ by $\mathbf{G}$ for the operation of $\mathbf{G}$ on $\mathbf{H}$ defined by $(g, h) \mapsto \text{Int}(u(g))h$ for $g \in \mathbf{G}(S)$, $h \in \mathbf{H}(S)$, $S \in \text{Ob}(\mathcal{C})$.

The set of these definitions and propositions carries over as usual to $\mathcal{C}$. One defines in particular the semi-direct product of two $\mathcal{C}$ -groups H and G (G operating on H) when the cartesian product $H \times G$ exists, and one has the analogue of proposition 2.3.7 in the following form:

⁽²⁵⁾**N.D.E.** Moreover, one says that $\mathbf{H}$ is *central* in $\mathbf{G}$ if $\text{Centr}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, what amounts to the same, if $\mathbf{H}(S)$ is central in $\mathbf{G}(S)$ for every S .

⁽²⁶⁾**N.D.E.** One has added definitions 2.3.6.1 and 2.3.6.2.

Proposition 2.3.8. *Let $H \xrightarrow{i} W \xrightarrow{f} G$ be a sequence of morphisms of $\mathcal{C}$ -groups such that for every $S \in \text{Ob}(\mathcal{C})$, $(H(S), i(S))$ is a kernel of $f(S): W(S) \rightarrow G(S)$. Let $u: G \rightarrow W$ be a morphism of $\mathcal{C}$ -groups which is a section of f . Then W is identified with the semi-direct product of H by G for the operation of G on H such that if $S \in \text{Ob}(\mathcal{C})$, if $g \in G(S)$ and $h \in H(S)$, one has $\text{Int}(u(g))i(h) = i(g^h)$.*

3. The category of $\mathbf{O}$ -modules, the category of $\mathbf{G}$ - $\mathbf{O}$ -modules

Definition 3.1. Let $\mathbf{O}$ and $\mathbf{F}$ be two objects of $\widehat{\mathcal{C}}$. One says that $\mathbf{F}$ is a $\widehat{\mathcal{C}}$ -module over the $\widehat{\mathcal{C}}$ -ring $\mathbf{O}$, or for short an $\mathbf{O}$ -module, if, for every $S \in \text{Ob}(\mathcal{C})$, one has endowed $\mathbf{O}(S)$ with a ring structure and $\mathbf{F}(S)$ with a module structure over this ring, in such a way that for every arrow $S' \rightarrow S''$ of $\mathcal{C}$, $\mathbf{O}(S'') \rightarrow \mathbf{O}(S')$ is a ring homomorphism and $\mathbf{F}(S'') \rightarrow \mathbf{F}(S')$ a homomorphism of abelian groups compatible with this ring homomorphism.

If $\mathbf{O}$ is fixed, one defines in the usual way a morphism of the $\mathbf{O}$ -modules $\mathbf{F}$ and $\mathbf{F}'$, whence the commutative group $\text{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and the category of $\mathbf{O}$ -modules, denoted $(\mathbf{O}\text{-Mod.})$.

Lemma 3.1.1. ⁽²⁷⁾ *$(\mathbf{O}\text{-Mod.})$ is endowed with the structure of an abelian category, defined “argument by argument”. Moreover, $(\mathbf{O}\text{-Mod.})$ satisfies axiom (AB 5) (cf. [Gr57, 1.5]), i.e. arbitrary direct sums exist and if $\mathbf{M}$ is an $\mathbf{O}$ -module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i \in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then*

$$\bigcup_{i \in I} (\mathbf{F}_i \cap \mathbf{N}) = \left(\bigcup_{i \in I} \mathbf{F}_i \right) \cap \mathbf{N}.$$

Indeed, let $f: \mathbf{F} \rightarrow \mathbf{F}'$ be a morphism of $\mathbf{O}$ -modules. One defines the $\mathbf{O}$ -modules $\text{Ker } f$ (resp. $\text{Im } f$ and $\text{Coker } f$) by setting, for every $S \in \text{Ob}(\mathcal{C})$, $(\text{Ker } f)(S) = \text{Ker } f(S)$ (resp. $\dots$). Then $\text{Ker } f$ (resp. $\text{Coker } f$) is a kernel (resp. a cokernel) of f , and one has an isomorphism of $\mathbf{O}$ -modules $\mathbf{F}/\text{Ker } f \xrightarrow{\sim} \text{Im } f$. This proves that $(\mathbf{O}\text{-Mod.})$ is an abelian category.

Arbitrary direct sums exist and are defined “argument by argument”. Finally, if $\mathbf{M}$ is an $\mathbf{O}$ -module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i \in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then the inclusion

$$\bigcup_{i \in I} (\mathbf{F}_i \cap \mathbf{N}) \subset \left(\bigcup_{i \in I} \mathbf{F}_i \right) \cap \mathbf{N}$$

is an equality: indeed, if $S \in \text{Ob}(\mathcal{C})$ and $x \in \mathbf{N}(S) \cap \bigcup_i \mathbf{F}_i(S)$, there exists $i \in I$ such that $x \in \mathbf{N}(S) \cap \mathbf{F}_i(S)$.

Proposition 3.1.2. ⁽²⁷⁾ *One assumes that the category $\mathcal{C}$ is small, i.e. that $\text{Ob}(\mathcal{C})$ is a set. Then $(\mathbf{O}\text{-Mod.})$ has a generator, and hence enough injective objects.*

PROOF. For every object S of $\mathcal{C}$, one defines the $\mathbf{O}$ -module $\mathbf{O}_S$ as follows. For every object S' , $\mathbf{O}_S(S')$ is a direct sum of copies of $\mathbf{O}(S')$ indexed by $\text{Hom}(S', S)$, i.e.

$$\mathbf{O}_S(S') = \bigoplus_{g: S' \rightarrow S} \mathbf{O}(S') g.$$

For every morphism $f: S'' \rightarrow S'$, denoting by f^* the ring homomorphism $\mathbf{O}(S') \rightarrow \mathbf{O}(S'')$, the morphism $\mathbf{O}_S(f): \mathbf{O}_S(S') \rightarrow \mathbf{O}_S(S'')$ sends each element ag of $\mathbf{O}_S(S')$ (where $a \in \mathbf{O}(S')$ and $g: S' \rightarrow S$) to the element $f^*(a)g \circ f$ of $\mathbf{O}_S(S'')$. One easily checks that this does indeed define a functor in $\mathbf{O}$ -modules.

Since, by hypothesis, $\text{Ob}(\mathcal{C})$ is a set, one may consider the direct sum $\mathbf{G} = \bigoplus_{S \in \text{Ob}(\mathcal{C})} \mathbf{O}_S$.

⁽²⁷⁾**N.D.E.** Statements 3.1.1 and 3.1.2 have been added in order to show that the category $(\mathbf{O}\text{-Mod.})$ is abelian and satisfies axiom (AB 5), and moreover has enough injective objects if the category $\mathcal{C}$ is small. An error in 3.1.2, pointed out in 2016 by Linyuan Liu, is corrected here.

Lemma 3.1.2.1. *Let $\mathbf{F}$ be an $\mathbf{O}$ -module. For every $S \in \text{Ob}(\mathcal{C})$, every element x of $\mathbf{F}(S)$ defines a unique morphism of $\mathbf{O}$ -modules $x: \mathbf{O}_S \rightarrow \mathbf{F}$ such that $x(1 \text{id}_S) = x$.*

PROOF. Let ϕ be a morphism of $\mathbf{O}$ -modules $\mathbf{O}_S \rightarrow \mathbf{F}$ such that $\phi(1 \text{id}_S) = x$. Let $g: S' \rightarrow S$ and $a \in \mathbf{O}(S')$. The $\mathbf{O}(S')$ -linearity and the commutativity of the diagram

$$\begin{array}{ccc} \mathbf{O}_S(S) & \xrightarrow{\phi} & \mathbf{F}(S) \\ \mathbf{O}_S(g) \downarrow & & \downarrow \mathbf{F}(g) \\ \mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \end{array}$$

imply that $\phi(ag) = a\phi(1g) = a\mathbf{F}(g)(\phi(1 \text{id}_S)) = a\mathbf{F}(g)(x)$. Thus ϕ , if it exists, is determined by the preceding equality. Conversely, if one defines ϕ in this way, then for every $f: S'' \rightarrow S'$ one has

$$\phi \circ \mathbf{O}_S(f)(ag) = \phi(f^*(a)g \circ f) = f^*(a)\mathbf{F}(g \circ f)(x) = \mathbf{F}(f)(a\mathbf{F}(g)(x)) = \mathbf{F}(f) \circ \phi(ag),$$

i.e. the diagram

$$\begin{array}{ccc} \mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \\ \mathbf{O}_S(f) \downarrow & & \downarrow \mathbf{F}(f) \\ \mathbf{O}_S(S'') & \xrightarrow{\phi} & \mathbf{F}(S'') \end{array}$$

is commutative. This proves the lemma. $\square$

Now let $\mathbf{F}$ be an $\mathbf{O}$ -module. For every $S \in \text{Ob}(\mathcal{C})$, let $F_0(S)$ be a system of generators of the $\mathbf{O}(S)$ -module $\mathbf{F}(S)$, and let $\mathbf{L}_S$ be the direct sum, indexed by $F_0(S)$, of copies of $\mathbf{O}_S$. By the lemma, one obtains a morphism of $\mathbf{O}$ -modules $\mathbf{L}_S \rightarrow \mathbf{F}$ which is such that the morphism $\mathbf{L}_S(S) \rightarrow \mathbf{F}(S)$ is surjective, hence an epimorphism of the category of $\mathbf{O}(S)$ -modules.

Set $I = \bigsqcup_{S \in \text{Ob}(\mathcal{C})} F_0(S)$. Then one obtains a morphism of $\mathbf{O}$ -modules

$$\mathbf{G}^{\oplus I} \rightarrow \bigoplus_{S \in \text{Ob}(\mathcal{C})} \mathbf{L}_S \rightarrow \mathbf{F}$$

which is an epimorphism “argument by argument”, hence an epimorphism of $(\mathbf{O}\text{-Mod.})$. This shows that $\mathbf{G}$ is a generator of $(\mathbf{O}\text{-Mod.})$ (cf. [Gr57, 1.9.1]). Since $(\mathbf{O}\text{-Mod.})$ satisfies (AB 5), it then follows from [Gr57, 1.10.1] that $(\mathbf{O}\text{-Mod.})$ has enough injective objects. $\square$

Remark 3.1.3. If $\mathbf{O}_0$ is the $\widehat{\mathcal{C}}$ -ring defined by $\mathbf{O}_0(S) = \mathbb{Z}$ (which must not be confused with the functor associated with the constant object $\mathbb{Z}$), then the category of $\mathbf{O}_0$ -modules is isomorphic to the category of commutative $\widehat{\mathcal{C}}$ -groups.

Definition 3.1.4. Let us observe that, if $\mathbf{F}$ is an $\mathbf{O}$ -module, then for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{F}_S$ is an $\mathbf{O}_S$ -module. One may therefore define an abelian $\widehat{\mathcal{C}}$ -group $\text{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$ by

$$\text{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')(S) = \text{Hom}_{\mathbf{O}_S}(\mathbf{F}_S, \mathbf{F}'_S).$$

One will likewise define objects

$$\text{Isom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}'), \quad \text{End}_{\mathbf{O}}(\mathbf{F}) \quad \text{and} \quad \text{Aut}_{\mathbf{O}}(\mathbf{F}),$$

the last being a $\widehat{\mathcal{C}}$ -group for the structure induced by the composition of automorphisms.

Definition 3.2. Let $\mathbf{O}$ be a $\widehat{\mathcal{C}}$ -ring, $\mathbf{F}$ an $\mathbf{O}$ -module, and $\mathbf{G}$ a $\widehat{\mathcal{C}}$ -group. A structure of $\mathbf{G}$ -object such that, for every $S \in \text{Ob}(\mathcal{C})$ and every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{F}(S)$ defined by g is an automorphism of its structure of $\mathbf{O}(S)$ -module, is called a *structure of $\mathbf{G}$ - $\mathbf{O}$ -module on $\mathbf{F}$* .

It amounts to the same to say that the morphism of $\widehat{\mathcal{C}}$ -groups

$$\rho: \mathbf{G} \rightarrow \text{Aut}(\mathbf{F})$$

defined in 2.3 sends $\mathbf{G}$ into the $\widehat{\mathcal{C}}$ -subgroup $\text{Aut}_{\mathbf{O}}(\mathbf{F})$ of $\text{Aut}(\mathbf{F})$. To give a *structure of $\mathbf{G}$ - $\mathbf{O}$ -module* on the $\mathbf{O}$ -module $\mathbf{F}$ is thus to give a *morphism of $\widehat{\mathcal{C}}$ -groups*

$$\rho: \mathbf{G} \longrightarrow \text{Aut}_{\mathbf{O}}(\mathbf{F}).$$

One defines in a natural way the abelian group $\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, hence the additive category of $\mathbf{G}$ - $\mathbf{O}$ -modules, denoted $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$.

Remark 3.2.1. The reader will observe that $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ may equally be defined as the category of $\mathbf{O}[\mathbf{G}]$ -modules, where $\mathbf{O}[\mathbf{G}]$ is the algebra of the $\widehat{\mathcal{C}}$ -group $\mathbf{G}$ over the $\widehat{\mathcal{C}}$ -ring $\mathbf{O}$, whose definition is clear.⁽²⁸⁾ Consequently, by 3.1.1 and 3.1.2, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ is an abelian category satisfying axiom (AB 5); moreover, if $\mathcal{C}$ is *small*, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects.

All the constructions of this paragraph specialize at once to the case where $\mathbf{G}$ (or $\mathbf{O}$, or both) are representable by objects of $\mathcal{C}$ which are thereby endowed with the corresponding algebraic structures.

We have treated succinctly the case of the principal algebraic structures encountered in the rest of this seminar. For the others (the structure of $\mathbf{O}$ -Lie algebra, for example), we believe that the reader will have had enough examples in this paragraph to be able, in each particular case, to put into operation for himself the general mechanism sketched in 2.2.

We are now going to apply what we have just done to the category of schemes, denoted (Sch) , and more generally to the categories $(\text{Sch})/S$ (which one will also denote (Sch/S)).

4. Algebraic structures in the category of schemes

We shall permit ourselves, whenever there is no ambiguity, the following abuses of language: one will denote by T the object $T \xrightarrow{f} S$ of $\mathcal{C}/S$, the datum of f ("structural morphism of T ") being understood, and one will identify $\mathcal{C}$ with a subcategory of $\widehat{\mathcal{C}}$. In view of the compatibilities stated in the preceding paragraphs, these identifications may be made without danger.

On the other hand, we shall simplify the terminology on the following model: a (Sch) -group will also be called a *group scheme*, a $(\text{Sch})/S$ -group a *group scheme over S* , or an *S -group scheme*, or an *S -group*, or an *A -group* when S is the spectrum of the ring A .

4.1. Constant schemes. — The category of schemes satisfies the conditions required in 1.8. One may therefore define constant objects there; for every set E , one has a scheme $E_{\mathbb{Z}}$ and, for every scheme S , an S -scheme $E_S = (E_{\mathbb{Z}})_S$. Let us recall (cf. *loc. cit.*) that for every S -scheme T , $\text{Hom}_S(T, E_S)$ is the set of locally constant maps from the topological space T to E .

The functor $E \mapsto E_S$ commutes with finite projective limits. In particular, if G is a group, G_S is an *S -group scheme*; if O is a ring, O_S is an *S -ring scheme*, etc.

4.2. Affine S -groups. — Let us recall a certain number of things about affine S -schemes (EGA II, § 1). One says that the S -scheme T is *affine over S* if the inverse image of every affine open of S is affine. The $\mathcal{O}_S$ -algebra $f_*(\mathcal{O}_T)$, which one denotes by $\mathcal{A}(T)$, is then *quasi-coherent* (f denotes the structural morphism of T). Conversely, to every quasi-coherent $\mathcal{O}_S$ -algebra $\mathcal{A}$, one may associate an S -scheme affine over S , denoted $\text{Spec}(\mathcal{A})$. These functors $T \mapsto \mathcal{A}(T)$ and $\mathcal{A} \mapsto \text{Spec}(\mathcal{A})$ are equivalences quasi-inverse to one another between the category of S -schemes affine over S and the category opposite to that of quasi-coherent $\mathcal{O}_S$ -algebras.

⁽²⁸⁾**N.D.E.** The following sentence has been added; this will be used in Section 5.

It follows that to give an algebraic structure on an S -scheme T affine over S is equivalent to giving the corresponding structure on $\mathcal{A}(T)$ in the category opposite to that of quasi-coherent $\mathcal{O}_S$ -algebras. In particular, if G is an affine S -group over S , $\mathcal{A}(G)$ is endowed with the structure of an augmented $\mathcal{O}_S$ -bialgebra, that is, one has two *morphisms of $\mathcal{O}_S$ -algebras*

$$\Delta: \mathcal{A}(G) \longrightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \quad \text{and} \quad \varepsilon: \mathcal{A}(G) \longrightarrow \mathcal{O}_S$$

corresponding to the morphisms of S -schemes

$$\pi: G \times G \longrightarrow G \quad \text{and} \quad e_G: S \longrightarrow G.$$

The maps Δ and ε satisfy the following conditions (which express that G is an S -monoid):⁽²⁹⁾

(HA 1) Δ is coassociative: the following diagram is commutative

$$\begin{array}{ccccc} & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) & & \\ & \Delta \nearrow & & \searrow \text{Id} \otimes \Delta & \\ \mathcal{A}(G) & & & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ & \Delta \searrow & & \nearrow \Delta \otimes \text{Id} & \\ & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) & & \end{array}$$

(HA 2): Δ is compatible with ε : the following two composites are the identity

$$\begin{aligned} \mathcal{A}(G) &\xrightarrow{\Delta} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\text{Id} \otimes \varepsilon} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathcal{A}(G), \\ \mathcal{A}(G) &\xrightarrow{\Delta} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\varepsilon \otimes \text{Id}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\sim} \mathcal{A}(G). \end{aligned}$$

Let us take the opportunity to remark once again that it follows from the definition of an S -group structure that, in order to give such a structure on an S -scheme G affine over S , it is not necessary to check anything at all on $\mathcal{A}(G)$, but simply to endow each $G(S')$, for S' over S , with a group structure functorial in S' . This remark applies *mutatis mutandis* to morphisms.

4.3. The groups $\mathbb{G}_a$ and $\mathbb{G}_m$. The ring $\mathbb{O}$

4.3.1. — Let $\mathbf{G}_a$ be the functor from $(\text{Sch})^\circ$ to (Ens) defined by

$$\mathbf{G}_a(S) = \Gamma(S, \mathcal{O}_S),$$

endowed with the structure of $(\widehat{\text{Sch}})$ -group defined by the additive group structure of the ring $\Gamma(S, \mathcal{O}_S)$. It is representable by an affine scheme which one will denote by $\mathbb{G}_a$, and which is therefore a *group scheme*

$$\mathbb{G}_a = \text{Spec } \mathbb{Z}[T].$$

Indeed, $\mathbf{G}_a(S) = \text{Hom}(S, \mathbb{G}_a) = \text{Hom}_{\text{Alg.}}(\mathbb{Z}[T], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S)$.

For every scheme S , one therefore has an affine S -group over S , denoted $\mathbb{G}_{a,S}$, which corresponds to the $\mathcal{O}_S$ -bialgebra $\mathcal{O}_S[T]$, with the diagonal map and the augmentation defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \quad \varepsilon(T) = 0.$$

4.3.2. — Let $\mathbf{G}_m$ be the functor from $(\text{Sch})^\circ$ to (Ens) defined by

$$\mathbf{G}_m(S) = \Gamma(S, \mathcal{O}_S)^\times,$$

⁽²⁹⁾**N.D.E.** And, of course, the inversion morphism $G \longrightarrow G$ induces a morphism of $\mathcal{O}_S$ -algebras $\tau: \mathcal{A}(G) \longrightarrow \mathcal{A}(G)$ which, together with Δ and ε , makes $\mathcal{A}(G)$ into a Hopf $\mathcal{O}_S$ -algebra.

where $\Gamma(S, \mathcal{O}_S)^\times$ denotes the multiplicative group of invertible elements of the ring $\Gamma(S, \mathcal{O}_S)$, endowed with its natural structure of $(\widehat{\mathbf{Sch}})$ -group. It is representable by an affine scheme denoted $\mathbb{G}_m$

$$\mathbb{G}_m = \mathrm{Spec} \mathbb{Z}[T, T^{-1}] = \mathrm{Spec} \mathbb{Z}[\mathbb{Z}],$$

where $\mathbb{Z}[\mathbb{Z}]$ denotes the algebra of the group $\mathbb{Z}$ over the ring $\mathbb{Z}$. Indeed

$$\mathbb{G}_m(S) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[T, T^{-1}], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S)^\times.$$

For every scheme S , one therefore has an S -group affine over S , denoted $\mathbb{G}_{m,S}$, which corresponds to the $\mathcal{O}_S$ -algebra $\mathcal{O}_S[\mathbb{Z}]$, with the diagonal map and the augmentation defined by:

$$\Delta(x) = x \otimes x \quad \text{and} \quad \varepsilon(x) = 1, \quad \text{for } x \in \mathbb{Z}.$$

4.3.3. — Let $\mathbf{O}$ be the functor $\mathbf{G}_a$ endowed with its structure of $(\widehat{\mathbf{Sch}})$ -ring. It is represented by the scheme $\mathrm{Spec} \mathbb{Z}[T]$, which one will denote by $\mathbb{O}$ when one considers it as endowed with its structure of *ring scheme*.

For every scheme S , $\mathbb{O}_S = S \times_{\mathrm{Spec} \mathbb{Z}} \mathrm{Spec} \mathbb{Z}[T] = \mathrm{Spec}(\mathcal{O}_S[T])$ is therefore a ring S -scheme, affine over S . (*Nota:* in EGA II 1.7.13, $\mathbb{O}_S$ is denoted by $S[T]$).

4.3.3.1. — ⁽³⁰⁾ For every object $\mathbf{X}$ of $(\widehat{\mathbf{Sch}})$, $\mathbf{O}(\mathbf{X}) = \mathrm{Hom}(\mathbf{X}, \mathbf{O})$ is endowed with a ring structure, functorial in $\mathbf{X}$. In particular, if X' is a scheme and if one is given morphisms $x: X' \rightarrow \mathbf{X}$ and $f: \mathbf{X} \rightarrow \mathbf{O}$ (i.e., $f \in \mathbf{O}(\mathbf{X})$), then $f(x) = f \circ x$ is an element of $\mathbf{O}(X') = \Gamma(X', \mathcal{O}_{X'})$.

Definition. — Let $\pi: \mathbf{M} \rightarrow \mathbf{X}$ be a morphism of $(\widehat{\mathbf{Sch}})$, and let $\mathbf{O}_\mathbf{X} = \mathbf{O} \times \mathbf{X}$. One says that $\mathbf{M}$ is an $\mathbf{O}_\mathbf{X}$ -module if one is given, for every $\mathbf{X}$ -scheme X' , a structure of $\mathbf{O}(X')$ -module on $\mathrm{Hom}_\mathbf{X}(X', \mathbf{M})$, functorial in X' .

This amounts to giving a law of abelian $\mathbf{X}$ -group $\mu: \mathbf{M} \times_\mathbf{X} \mathbf{M} \rightarrow \mathbf{M}$ and an “external law”

$$\mathbf{O} \times \mathbf{M} = \mathbf{O}_\mathbf{X} \times_\mathbf{X} \mathbf{M} \rightarrow \mathbf{M}, \quad (f, m) \mapsto f \cdot m$$

which is an $\mathbf{X}$ -morphism (i.e. $\pi(f \cdot m) = \pi(m)$) and which, for every $x \in \mathbf{X}(S)$, endows $\mathbf{M}(x) = \{m \in \mathbf{M}(S) \mid \pi(m) = x\}$ with a structure of $\mathbf{O}(S)$ -module.

In this case, for every $\mathbf{Y} \in \mathrm{Ob}(\widehat{\mathbf{Sch}})_\mathbf{X}$ (not necessarily representable), $\mathrm{Hom}_\mathbf{X}(\mathbf{Y}, \mathbf{M}) = \Gamma(\mathbf{M}_\mathbf{Y}/\mathbf{Y})$ is an $\mathbf{O}(\mathbf{Y})$ -module, functorially in $\mathbf{Y}$.

4.4. Diagonalizable groups

4.4.1. — The construction of $\mathbb{G}_m$ generalizes as follows: let M be a commutative group and $M_\mathbb{Z}$ the group scheme associated with it by the procedure of 4.1. Consider the functor defined by

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{groups}}(M, \mathbb{G}_m(S)) \simeq \mathrm{Hom}_{S\text{-Gr.}}(M_S, \mathbb{G}_{m,S}).$$

It is a commutative $(\widehat{\mathbf{Sch}})$ -group which is representable by a *group scheme* which one will denote by $\mathbf{D}(M)$; one will therefore have by definition:

$$\mathbf{D}(M) \simeq \mathrm{Hom}_{(\mathrm{Sch})\text{-Gr.}}(M_\mathbb{Z}, \mathbb{G}_m).$$

Indeed, set

$$\mathbf{D}(M) = \mathrm{Spec} \mathbb{Z}[M],$$

where $\mathbb{Z}[M]$ is the algebra of the group M over the ring $\mathbb{Z}$; one has

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[M], \Gamma(S, \mathcal{O}_S)) \simeq \mathrm{Hom}_{\mathrm{groups}}(M, \Gamma(S, \mathcal{O}_S)^\times)$$

by the very definition of the algebra $\mathbb{Z}[M]$.

⁽³⁰⁾**N.D.E.** This paragraph has been added; it will be useful later (cf. II 1.3).

4.4.2. — For every scheme S one therefore has an S -group scheme affine over S

$$\mathbf{D}_S(M) = \mathbf{D}(M)_S = \mathrm{Hom}_{(\mathrm{Sch})\text{-}\mathrm{Gr.}}(M_{\mathbb{Z}}, \mathbb{G}_m)_S = \mathrm{Hom}_{S\text{-}\mathrm{Gr.}}(M_S, \mathbb{G}_{m,S}).$$

It is associated with the $\mathcal{O}_S$ -bialgebra $\mathcal{O}_S[M]$, equipped with the diagonal map and the augmentation defined by

$$\Delta(x) = x \otimes x \quad \text{and} \quad \varepsilon(x) = 1, \quad \text{for } x \in M.$$

4.4.3. — If $f: M \rightarrow N$ is a homomorphism of commutative groups, one defines in an evident way a morphism of S -groups

$$\mathbf{D}_S(f): \mathbf{D}_S(N) \rightarrow \mathbf{D}_S(M),$$

whence a *functor*

$$\mathbf{D}_S: M \mapsto \mathbf{D}_S(M)$$

from the category of abelian groups to the category of affine S -groups over S , which one can also define as the composite of the functor $M \mapsto M_S$ and of the functor $M_S \mapsto \mathrm{Hom}_{S\text{-}\mathrm{Gr.}}(M_S, \mathbb{G}_{m,S})$. This functor *commutes with extensions of the base*.

An S -group isomorphic to a group of the form $\mathbf{D}_S(M)$ is said to be *diagonalizable*. Let us note that the elements of M are interpreted as certain characters of $\mathbf{D}_S(M)$, that is, certain elements of $\mathrm{Hom}_{S\text{-}\mathrm{Gr.}}(\mathbf{D}_S(M), \mathbb{G}_{m,S})$. (*Confer* VIII 1).

4.4.4. — Let us give some examples of diagonalizable groups. First, one has

$$\mathbf{D}(\mathbb{Z}) = \mathbb{G}_m \quad \text{and} \quad \mathbf{D}(\mathbb{Z}^r) = (\mathbb{G}_m)^r.$$

One sets

$$\mu_n = \mathbf{D}(\mathbb{Z}/n\mathbb{Z}),$$

and one calls it the group of n -th roots of unity. Indeed, one has

$$\mu_n(S) = \mathrm{Hom}_{\mathrm{groups}}(\mathbb{Z}/n\mathbb{Z}, \Gamma(S, \mathcal{O}_S)^\times) = \{f \in \Gamma(S, \mathcal{O}_S) \mid f^n = 1\}.$$

The S -group $\mu_{n,S}$ corresponds to the $\mathcal{O}_S$ -algebra $\mathcal{O}_S[T]/(T^n - 1)$. Suppose in particular that S is the spectrum of a field k of characteristic $p = n$. Setting $T - 1 = s$, one finds

$$k[T]/(T^p - 1) = k[s]/(s^p),$$

which shows that the underlying topological space of $\mu_{p,k}$ is reduced to a point, the local ring of this point being the artinian k -algebra $k[s]/(s^p)$. (In the same vein, let us point out that the S -schemes $\mathbb{G}_{a,S}$, $\mathbb{G}_{m,S}$, $\mathcal{O}_S$ are *smooth* over S , that $\mathbf{D}_S(M)$ is flat over S and that it is formally smooth (resp. *smooth*) over S if and only if no residual characteristic of S divides the torsion of M (resp. and if moreover M is of finite type), cf. Exp. VIII, 2.1).

4.5. Other examples of groups. — The preceding procedure applies to the “classical groups” (linear groups $\mathbf{GL}_n$, symplectic groups $\mathbf{Sp}_n$, orthogonal groups $\mathbf{O}_n$, etc.). One will define for example $\mathbf{GL}_n$ as representing the functor $\mathbf{GL}_n$ such that:

$$\mathbf{GL}_n(S) = \mathbf{GL}(n, \Gamma(S, \mathcal{O}_S)) = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{O}_S^n).$$

One can construct it for example as the open subscheme of $\mathrm{Spec} \mathbb{Z}[T_{ij}]$ ($1 \leq i, j \leq n$) defined by the function $f = \det((T_{ij})_{i,j=1}^n)$, that is, $\mathrm{Spec} \mathbb{Z}[T_{ij}, f^{-1}]$.

4.6. Module-functors in the category of schemes. — We propose to associate with every $\mathcal{O}_S$ -module on the scheme S , an $\mathbf{O}_S$ -module (where $\mathbf{O}_S$ denotes the ring-functor introduced in 4.3.3). This can be done in two different ways. To be precise:

Definition 4.6.1. — Let S be a scheme. For every $\mathcal{O}_S$ -module $\mathcal{F}$ one denotes by $\mathbf{V}(\mathcal{F})$ and $\mathbf{W}(\mathcal{F})$ the contravariant functors on $(\mathrm{Sch})_S$ defined by:

$$\begin{aligned} \mathbf{V}(\mathcal{F})(S') &= \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}, \mathcal{O}_{S'}), \\ \mathbf{W}(\mathcal{F})(S') &= \Gamma(S', \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \end{aligned}$$

(where $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ denotes the inverse image over S' of the $\mathcal{O}_S$ -module $\mathcal{F}$).

Then $\mathbf{V}(\mathcal{F})$ and $\mathbf{W}(\mathcal{F})$ are endowed in an evident way with a structure of $\mathbf{O}_S$ -modules (one recalls that $\mathbf{O}_S(S') = \Gamma(S', \mathcal{O}_{S'}) = \mathbf{W}(\mathcal{O}_S)(S')$), in such a way that one has in fact defined two functors $\mathbf{V}$ and $\mathbf{W}$ from the category of $\mathcal{O}_S$ -modules to the category of $\mathbf{O}_S$ -modules, $\mathbf{V}$ being contravariant and $\mathbf{W}$ covariant.

We restrict ourselves in the rest of this paragraph to the case in which the $\mathcal{O}_S$ -modules in question are quasi-coherent, that is, we consider $\mathbf{V}$ and $\mathbf{W}$ as functors from the category ($\mathcal{O}_S\text{-Mod.q.c.}$) of quasi-coherent $\mathcal{O}_S$ -modules to the category of $\mathbf{O}_S$ -modules

$$\begin{aligned} \mathbf{V}: (\mathcal{O}_S\text{-Mod.q.c.})^\circ &\longrightarrow (\mathbf{O}_S\text{-Mod.}), \\ \mathbf{W}: (\mathcal{O}_S\text{-Mod.q.c.}) &\longrightarrow (\mathbf{O}_S\text{-Mod.}). \end{aligned}$$

Remark 4.6.1.1. — ⁽³¹⁾ The reader will notice that, in the sequel, all the propositions which involve only the functor $\mathbf{W}$ are valid for arbitrary $\mathcal{O}_S$ -modules, not necessarily quasi-coherent.

Proposition 4.6.2. —

- (i) *The functors $\mathbf{V}$ and $\mathbf{W}$ commute with extension of the base: if S' is over S and if $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_S$ -module, one has*

$$\mathbf{V}(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \simeq \mathbf{V}(\mathcal{F})_{S'} \quad \text{and} \quad \mathbf{W}(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \simeq \mathbf{W}(\mathcal{F})_{S'}.$$

- (ii) *The functors $\mathbf{V}$ and $\mathbf{W}$ are fully faithful: the canonical maps*

$$\begin{aligned} \text{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') &\longrightarrow \text{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F})) \\ \text{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') &\longrightarrow \text{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}')) \end{aligned}$$

are bijective.

- (iii) *The functors $\mathbf{V}$ and $\mathbf{W}$ are additive:*

$$\mathbf{V}(\mathcal{F} \oplus \mathcal{F}') \simeq \mathbf{V}(\mathcal{F}) \times_S \mathbf{V}(\mathcal{F}') \quad \text{and} \quad \mathbf{W}(\mathcal{F} \oplus \mathcal{F}') \simeq \mathbf{W}(\mathcal{F}) \times_S \mathbf{W}(\mathcal{F}').$$

Parts (i) and (iii) are evident from the definitions. For (ii), one takes for S' opens of S . We leave the proof to the reader (for $\mathbf{V}$, use EGA II, 1.7.14).

Let us recall (cf. 3.1.4) that if $\mathcal{F}, \mathcal{F}'$ are $\mathcal{O}_S$ -modules, $\text{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}')$ denotes the S -functor (in abelian groups) which to every $S' \rightarrow S$ associates $\text{Hom}_{\mathcal{O}_{S'}}(\mathcal{F}_{S'}, \mathcal{F}'_{S'})$.

Proposition 4.6.3. — ⁽³²⁾ *One has canonical morphisms in $(\mathbf{O}_S\text{-Mod.})$:*

$$\begin{array}{ccc} \text{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}')) & & \text{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F})) \\ & \nwarrow \quad \nearrow & \\ & \mathbf{W}(\text{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}')) & \end{array}$$

This follows immediately from 4.6.2 (i) and (ii).

Notation 4.6.3.1. — Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$ -module. One knows (EGA II, 1.7.8) that the S -functor $\mathbf{V}(\mathcal{F})$ is representable by an S -scheme affine over S , which one denotes by $\mathbf{V}(\mathcal{F})$ and which one calls the vector fibration⁽³³⁾ defined by $\mathcal{F}$:

$$\mathbf{V}(\mathcal{F}) = \text{Spec}(\mathcal{S}(\mathcal{F})),$$

⁽³¹⁾**N.D.E.** The numbering 4.6.1.1 has been added, for later references.

⁽³²⁾**N.D.E.** The isomorphism $\text{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}')) \simeq \text{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F}))$ has been added.

⁽³³⁾**N.D.E.** “vector bundle” has been replaced by “vector fibration”; current usage being to call “vector bundle of rank r ” a vector fibration which is locally trivial of rank r , i.e., whose sheaf of sections is locally isomorphic to $\mathcal{O}_S^{\oplus r}$.

where $\mathcal{S}(\mathcal{F})$ denotes the symmetric algebra of the $\mathcal{O}_S$ -module $\mathcal{F}$.⁽³⁴⁾

Proposition 4.6.4. — *Let $\mathcal{F}$ and $\mathcal{F}'$ be two quasi-coherent $\mathcal{O}_S$ -modules, and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_S$ -algebra. One has a functorial isomorphism:*

$$\mathrm{Hom}_S(\mathrm{Spec}(\mathcal{A}), \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}'), \mathbf{W}(\mathcal{F}))) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}', \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}).$$

Indeed, if one denotes $X = \mathrm{Spec}(\mathcal{A})$, the first member is canonically isomorphic to $\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}'), \mathbf{W}(\mathcal{F}))(X)$, that is, by definition, to

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{F}')_X, \mathbf{W}(\mathcal{F})_X) \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{F}' \otimes \mathcal{O}_X), \mathbf{W}(\mathcal{F} \otimes \mathcal{O}_X))$$

(cf. 4.6.2 (i)), which by 4.6.2 (ii) may also be written

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}' \otimes \mathcal{O}_X, \mathcal{F} \otimes \mathcal{O}_X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}', \pi_*(\pi^*(\mathcal{F}))),$$

where one denotes by $\pi: X \rightarrow S$ the structural morphism. But, by EGA II, 1.4.7, one has $\pi_*(\pi^*(\mathcal{F})) \simeq \mathcal{F} \otimes \mathcal{A}$, which completes the proof.

Corollary 4.6.4.1. — *One has a canonical isomorphism*

$$\mathbf{W}(\mathcal{F} \otimes \mathcal{A}) \simeq \mathrm{Hom}_S(\mathrm{Spec}(\mathcal{A}), \mathbf{W}(\mathcal{F})).$$

Indeed,⁽³⁵⁾ let $f: S' \rightarrow S$ be an S -scheme and $X' = X \times_S S'$; one has a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & X \\ \pi' \downarrow & & \downarrow \pi \\ S' & \xrightarrow{f} & S \end{array}$$

and by EGA II, 1.5.2, X' is affine over S' and $\pi'_*(\mathcal{O}_{X'}) = f^*(\mathcal{A})$. One therefore has

$$\mathrm{Hom}_S(\mathrm{Spec}(\mathcal{A}), \mathbf{W}(\mathcal{F}))(S') = \mathrm{Hom}_{S'}(\mathrm{Spec}(f^*(\mathcal{A})), \mathbf{W}(f^*(\mathcal{F})))$$

and by 4.6.4 applied to $f^*(\mathcal{F})$, $\mathcal{F}' = \mathcal{O}_{S'}$ and $f^*(\mathcal{A})$, this equals

$$\Gamma(S', f^*(\mathcal{F}) \otimes f^*(\mathcal{A})) = \Gamma(S', f^*(\mathcal{F} \otimes \mathcal{A})) = \mathbf{W}(\mathcal{F} \otimes \mathcal{A})(S').$$

Proposition 4.6.5. — *If $\mathcal{F}$ and $\mathcal{F}'$ are two locally free $\mathcal{O}_S$ -modules of finite type, the morphisms of 4.6.3 are isomorphisms.*

Indeed, for every $S' \rightarrow S$, one has

$$\begin{aligned} \mathbf{W}(\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}'))(S') &= \Gamma(S', \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') \otimes \mathcal{O}_{S'}) \\ &= \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{F} \otimes \mathcal{O}_{S'}, \mathcal{F}' \otimes \mathcal{O}_{S'}). \end{aligned}$$

But the second member is indeed isomorphic to

$$\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}'))(S')$$

and to

$$\mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F}))(S'),$$

by 4.6.2 (i) and (ii).

Corollary 4.6.5.1. — *Let $\mathcal{F}$ be a locally free $\mathcal{O}_S$ -module of finite type. Set*

$$\mathcal{F}^\vee = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{O}_S).$$

⁽³⁴⁾**N.D.E.** Let us point out here the articles [Ni04] (resp. [Ni02]) which show that if S is noetherian and $\mathcal{F}$ is a coherent $\mathcal{O}_S$ -module, then $\mathbf{W}(\mathcal{F})$ (resp. the S -group which to every $T \rightarrow S$ associates $\mathrm{Aut}_{\mathcal{O}_T}(\mathcal{F} \otimes \mathcal{O}_T)$) is representable if and only if $\mathcal{F}$ is locally free.

⁽³⁵⁾**N.D.E.** The original has been spelled out in what follows.

One has canonical isomorphisms:

$$\begin{aligned}\mathbf{W}(\mathcal{F}^\vee) &\simeq \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{O}_S) \simeq \mathbf{V}(\mathcal{F}), \\ \mathbf{V}(\mathcal{F}^\vee) &\simeq \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}), \mathbf{O}_S) \simeq \mathbf{W}(\mathcal{F}).\end{aligned}$$

Finally, one has the following proposition:

Proposition 4.6.6. — *Let $f: \mathcal{F} \rightarrow \mathcal{F}'$ be a morphism of locally free $\mathcal{O}_S$ -modules of finite rank. In order that $\mathbf{W}(f): \mathbf{W}(\mathcal{F}) \rightarrow \mathbf{W}(\mathcal{F}')$ be a monomorphism, it is necessary and sufficient that f identify $\mathcal{F}$ locally with a direct factor of $\mathcal{F}'$.*

The direct statement is essentially contained in EGA 0_I, 5.5.5.⁽³⁶⁾ Conversely, if $\mathcal{F}$ is locally a direct factor of $\mathcal{F}'$ then, for every $\pi: S' \rightarrow S$, $\pi^*\mathcal{F}$ is a submodule of $\pi^*\mathcal{F}'$ (since it is locally a direct factor), hence $\mathbf{W}(\mathcal{F})(S') = \Gamma(S', \pi^*\mathcal{F})$ is a submodule of $\mathbf{W}(\mathcal{F}')(S') = \Gamma(S', \pi^*\mathcal{F}')$.

4.7. The category of G - $\mathcal{O}_S$ -modules. — Let G be an S -group and $\mathcal{F}$ an $\mathcal{O}_S$ -module; $\mathbf{W}(\mathcal{F})$ is endowed with a structure of $\mathbf{O}_S$ -module.

Definition 4.7.1. — One calls a structure of G - $\mathcal{O}_S$ -module on $\mathcal{F}$ a structure of h_G - $\mathbf{O}_S$ -module on $\mathbf{W}(\mathcal{F})$ (cf. 3.2). A morphism of G - $\mathcal{O}_S$ -modules is by definition a morphism of the associated h_G - $\mathbf{O}_S$ -modules. One thus obtains the category $(G\text{-}\mathcal{O}_S\text{-Mod.})$, and one denotes by $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ the full subcategory formed by the G - $\mathcal{O}_S$ -modules which are quasi-coherent as $\mathcal{O}_S$ -modules.

To give a structure of G - $\mathcal{O}_S$ -module on $\mathcal{F}$ is therefore, by 3.2, to give a morphism of $(\widehat{\mathrm{Sch}})_{/S}$ -groups

$$\rho: h_G \rightarrow \mathrm{Aut}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F})).$$

Remark 4.7.1.0. — ⁽³⁷⁾ Since one has, by 4.6.2, an *anti-isomorphism* of S -group functors

$$(\dagger) \quad \mathrm{Aut}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F})) \simeq \mathrm{Aut}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F})),$$

one sees that it is equivalent to give a structure of h_G - $\mathbf{O}_S$ -module on $\mathbf{W}(\mathcal{F})$ or on $\mathbf{V}(\mathcal{F})$. Indeed, let $\rho: h_G \rightarrow \mathrm{Aut}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}))$ be as above. For every $T \rightarrow S$ and $g \in G(T)$, denote by $\rho^*(g)$ the image of $\rho(g)$ under the anti-isomorphism $(\dagger)$; one therefore has $\rho^*(gh) = \rho^*(h) \circ \rho^*(g)$, i.e. ρ^* defines a structure of h_G - $\mathbf{O}_S$ -module “on the right” on $\mathbf{V}(\mathcal{F})$. Setting $\rho^\vee(g) = \rho^*(g^{-1})$, one indeed obtains a structure of h_G - $\mathbf{O}_S$ -module on $\mathbf{V}(\mathcal{F})$, the datum of which is equivalent to that of ρ .

Remark 4.7.1.1. — One may say that the categories which one has just defined have been constructed by the cartesian squares:

$$\begin{array}{ccccc}(G\text{-}\mathcal{O}_S\text{-Mod.q.c.}) & \hookrightarrow & (G\text{-}\mathcal{O}_S\text{-Mod.}) & \hookrightarrow & (h_G\text{-}\mathbf{O}_S\text{-Mod.}) \\ \downarrow & & \downarrow & & \downarrow \text{forgetful} \\ (\mathcal{O}_S\text{-Mod.q.c.}) & \hookrightarrow & (\mathcal{O}_S\text{-Mod.}) & \xrightarrow{\mathbf{W}} & (\mathbf{O}_S\text{-Mod.})\end{array}$$

⁽³⁸⁾ The categories $(\mathcal{O}_S\text{-Mod.})$ and $(\mathbf{O}_S\text{-Mod.})$ are abelian, but one must take care that in general the functor $\mathbf{W}$ is not exact, neither on the left nor on the right.

Remark 4.7.1.2. — ⁽³⁹⁾ Let $\mathcal{F}$ be a G - $\mathcal{O}_S$ -module. The subsheaf of invariants $\mathcal{F}^G$ is defined as follows: for every open U of S ,

$$\mathcal{F}^G(U) = \mathbf{W}(\mathcal{F})^G(U) = \{x \in \mathcal{F}(U) \mid g \cdot x_{S'} = x_{S'} \text{ for every } S' \xrightarrow{f} U, g \in G(S')\},$$

⁽³⁶⁾**N.D.E.** The following sentence has been spelled out.

⁽³⁷⁾**N.D.E.** This remark has been added, taken from [DG], II, §2, 1.1.

⁽³⁸⁾**N.D.E.** The original has been corrected by removing the assertion that the category $(G\text{-}\mathcal{O}_S\text{-Mod.})$ is abelian; see 4.7.2.1 below.

⁽³⁹⁾**N.D.E.** This remark has been added.

where $x_{S'}$ denotes the image of x in $\Gamma(S', f^*(\mathcal{F})) = \Gamma(U, f_* f^*(\mathcal{F}))$.

One must take care that the natural morphism $\mathbf{W}(\mathcal{F}^G) \rightarrow \mathbf{W}(\mathcal{F})^G$ is not an isomorphism in general. For example, if $S = \text{Spec}(\mathbb{Z})$ and G is the constant group $\mathbb{Z}/2\mathbb{Z} = \{1, \tau\}$ acting on $\mathcal{F} = \mathcal{O}_S$ by $\tau \cdot 1 = -1$, one has $\mathcal{F}^G = 0$ but, if R is an $\mathbb{F}_2$ -algebra, $\mathbf{W}(\mathcal{F})^G(\text{Spec}(R)) = R$.

4.7.2. — One assumes henceforth, until the end of no. 4.7, that G is *affine* over S .⁽⁴⁰⁾ Then, by virtue of 4.6.4, the datum of a morphism of S -functors

$$\rho: h_G \rightarrow \underline{\text{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{F}))$$

is equivalent to that of a morphism of $\mathcal{O}_S$ -modules

$$\mu: \mathcal{F} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

To say that ρ is a morphism of $(\widehat{\mathbf{Sch}})_{/S}$ -groups is then equivalent to saying that μ satisfies the following axioms:

(CM 1) *the following diagram is commutative*

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\mu} & \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ \mu \downarrow & & \downarrow \text{id} \otimes \Delta \\ \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) & \xrightarrow{\mu \otimes \text{id}} & \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G). \end{array}$$

(CM 2) *the composite below is the identity*

$$\mathcal{F} \xrightarrow{\mu} \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\text{id} \otimes \varepsilon} \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathcal{F}.$$

These axioms (CM 1) and (CM 2) are those of the structure of a (right) comodule⁽⁴¹⁾ over the bialgebra $\mathcal{A}(G)$.

Set $\mathcal{A} = \mathcal{A}(G)$. If $\mathcal{F}$ and $\mathcal{F}'$ are $\mathcal{A}$ -comodules, a morphism of comodules $f: \mathcal{F} \rightarrow \mathcal{F}'$ is a morphism of $\mathcal{O}_S$ -modules such that the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{f} & \mathcal{F}' \\ \mu_{\mathcal{F}} \downarrow & & \downarrow \mu_{\mathcal{F}'} \\ \mathcal{F} \otimes \mathcal{A} & \xrightarrow{f \otimes \text{id}} & \mathcal{F}' \otimes \mathcal{A}. \end{array}$$

One thus obtains the category $(\mathcal{A}\text{-Comod.})$, and one will denote by $(\mathcal{A}\text{-Comod.q.c.})$ the full subcategory consisting of the $\mathcal{A}$ -comodules which are quasi-coherent as $\mathcal{O}_S$ -modules. One has thus obtained:

Proposition 4.7.2. — *Let G be an S -group affine over S . One has equivalences of categories:*

$$(G\text{-}\mathcal{O}_S\text{-Mod.}) \simeq (\mathcal{A}(G)\text{-Comod.})$$

$$(G\text{-}\mathcal{O}_S\text{-Mod.q.c.}) \simeq (\mathcal{A}(G)\text{-Comod.q.c.}).$$

⁽⁴²⁾ *If moreover $S = \text{Spec}(\Lambda)$ is affine and if one denotes $\Lambda[G] = \Gamma(S, \mathcal{A}(G))$, one has an equivalence of categories*

$$(\mathcal{A}(G)\text{-Comod.q.c.}) \simeq (\Lambda[G]\text{-Comod.}).$$

⁽⁴³⁾ Suppose moreover that $\mathcal{A} = \mathcal{A}(G)$ is a *flat* $\mathcal{O}_S$ -module. Let $\mathcal{E}$ be an $\mathcal{A}$ -comodule and $\mathcal{F}$ a sub- $\mathcal{O}_S$ -module of $\mathcal{E}$. Since $\mathcal{A}$ is flat over $\mathcal{O}_S$, one can identify $\mathcal{F} \otimes \mathcal{A}$ (resp. $\mathcal{F} \otimes \mathcal{A} \otimes \mathcal{A}$) with a sub- $\mathcal{O}_S$ -module of $\mathcal{E} \otimes \mathcal{A}$ (resp. $\mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A}$). Suppose that $\mu_{\mathcal{E}}$ maps $\mathcal{F}$ into $\mathcal{F} \otimes \mathcal{A}$;

⁽⁴⁰⁾ **N.D.E.** Cf. VI_B, §§ 11.1–11.6 for the extension of the results of 4.7.2 to the case where G is not necessarily affine, but where G and $\mathcal{F}$ are assumed *flat* over S .

⁽⁴¹⁾ **N.D.E.** Left $G\text{-}\mathcal{O}_S$ -modules correspond naturally to right $\mathcal{A}(G)$ -comodules.

⁽⁴²⁾ **N.D.E.** The following sentence has been added.

⁽⁴³⁾ **N.D.E.** The following paragraph has been added, taken from [Se68, § 1.3].

then its restriction $\mu_{\mathcal{F}}: \mathcal{F} \rightarrow \mathcal{F} \otimes \mathcal{A}$ equips $\mathcal{F}$ with a structure of $\mathcal{A}$ -comodule; one says that $\mathcal{F}$ is a *subcomodule* of $\mathcal{E}$. By passage to the quotient, $\mu_{\mathcal{E}}$ defines a morphism of $\mathcal{O}_S$ -modules $\mathcal{E}/\mathcal{F} \rightarrow \mathcal{E}/\mathcal{F} \otimes \mathcal{A}$, which equips $\mathcal{E}/\mathcal{F}$ with a structure of $\mathcal{A}$ -comodule. If $f: \mathcal{E} \rightarrow \mathcal{E}'$ is a morphism of $\mathcal{A}$ -comodules, $\text{Ker } f$ (resp. $\text{Im } f$) is a sub- $\mathcal{A}$ -comodule of $\mathcal{E}$ (resp. $\mathcal{E}'$), and f induces an isomorphism of $\mathcal{A}$ -comodules: $\mathcal{E}/\text{Ker } f \xrightarrow{\sim} \text{Im } f$. Moreover, if $\mathcal{E}$ and $\mathcal{E}'$ are quasi-coherent $\mathcal{O}_S$ -modules, the same is true of $\text{Ker } f$ and $\text{Im } f$. Consequently, $(\mathcal{A}\text{-Comod.})$ and $(\mathcal{A}\text{-Comod.q.c.})$ are *abelian* categories.

Corollary 4.7.2.1. — *One assumes that G is affine and flat over S . Then the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ (resp. $(G\text{-}\mathcal{O}_S\text{-Mod.})$), equivalent to the category of quasi-coherent $\mathcal{A}(G)$ -comodules over $\mathcal{O}_S$ (resp. of $\mathcal{A}(G)$ -comodules), is abelian.*

4.7.3. — Suppose now that G is a *diagonalizable* group, that is, that $\mathcal{A}(G)$ is the algebra of a commutative group M over the sheaf of rings $\mathcal{O}_S$. If $\mathcal{F}$ is an $\mathcal{O}_S$ -module, one has

$$\mathcal{F} \otimes \mathcal{A}(G) = \coprod_{m \in M} \mathcal{F} \otimes m\mathcal{O}_S.$$

To give a morphism of $\mathcal{O}_S$ -modules

$$\mu: \mathcal{F} \rightarrow \mathcal{F} \otimes \mathcal{A}(G)$$

is therefore equivalent to giving $\mathcal{O}_S$ -endomorphisms $(\mu_m)_{m \in M}$ of $\mathcal{F}$, such that for every section x of $\mathcal{F}$ over an open of S , $(\mu_m(x))$ is a section of the direct sum $\coprod_{m \in M} \mathcal{F}$ (this means that over every sufficiently small open, only finitely many of the restrictions of the $\mu_m(x)$ are nonzero).

In order that μ , defined by

$$\mu(x) = \sum_{m \in M} \mu_m(x) \otimes m,$$

satisfy (CM 1) (resp. (CM 2)), it is necessary and sufficient that one have

$$\mu_m \circ \mu_{m'} = \delta_{mm'} \mu_m, \quad \left(\text{resp. } \sum_{m \in M} \mu_m = \text{Id}_{\mathcal{F}} \right),$$

which means that the μ_m are pairwise orthogonal projectors summing to the identity. One has thus proved:

Proposition 4.7.3. — *If $G = \mathbf{D}_S(M)$ is a diagonalizable S -group, the category of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules (resp. of $G\text{-}\mathcal{O}_S$ -modules) is equivalent to the category of quasi-coherent $\mathcal{O}_S$ -modules (resp. of $\mathcal{O}_S$ -modules) graded of type M .*

Remark. — If $\mathcal{F}$ is endowed with a structure of $\mathcal{O}_S$ -algebra preserved by the operations of G , then the grading of $\mathcal{F}$ is an algebra grading. More precisely:

Corollary 4.7.3.1. — *The functor $\mathcal{A} \mapsto \text{Spec } \mathcal{A}$ induces an equivalence between the category of quasi-coherent $\mathcal{O}_S$ -algebras graded of type M and the opposite category to that of the S -schemes affine over S with S -group of operators $G = \mathbf{D}_S(M)$.*

Proposition 4.7.4. — *Let G be a diagonalizable S -group. If $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$ is an exact sequence of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules which splits as a sequence of $\mathcal{O}_S$ -modules, then it likewise splits as a sequence of $G\text{-}\mathcal{O}_S$ -modules.*

Indeed, if $G = \mathbf{D}_S(M)$, each of the $\mathcal{F}_i$ is graded by the $(\mathcal{F}_i)_m$ and for each $m \in M$ the sequence

$$0 \rightarrow (\mathcal{F}_1)_m \rightarrow (\mathcal{F}_2)_m \rightarrow (\mathcal{F}_3)_m \rightarrow 0$$

of $\mathcal{O}_S$ -modules is split. The preceding proposition then yields the result.

5. Cohomology of groups

5.1. The standard complex. — ⁽⁴⁴⁾Let $\mathcal{C}$ be a category, $\mathbf{G}$ a $\widehat{\mathcal{C}}$ -group, $\mathbf{O}$ a $\widehat{\mathcal{C}}$ -ring and $\mathbf{F}$ a $\mathbf{G}$ - $\mathbf{O}$ -module. One sets, for $n \geq 0$,

$$\underline{C}^n(\mathbf{G}, \mathbf{F}) = \underline{\text{Hom}}(\mathbf{G}^n, \mathbf{F}) \quad \text{and} \quad C^n(\mathbf{G}, \mathbf{F}) = \text{Hom}(\mathbf{G}^n, \mathbf{F}),$$

where $\mathbf{G}^0$ is the final object e . Then $\underline{C}^n(\mathbf{G}, \mathbf{F})$ (resp. $C^n(\mathbf{G}, \mathbf{F})$) is endowed in an evident way with a structure of $\mathbf{O}$ -module (resp. of $\Gamma(\mathbf{O})$ -module), and one has

$$C^n(\mathbf{G}, \mathbf{F}) \simeq \Gamma(\underline{C}^n(\mathbf{G}, \mathbf{F})) \quad \text{and} \quad \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S).$$

To give an element of $C^n(\mathbf{G}, \mathbf{F})$ is to give, for each $S \in \text{Ob}(\mathcal{C})$, an n -cochain of $\mathbf{G}(S)$ in $\mathbf{F}(S)$, functorially in S . The boundary operator

$$\partial: C^n(\mathbf{G}(S), \mathbf{F}(S)) \longrightarrow C^{n+1}(\mathbf{G}(S), \mathbf{F}(S)),$$

which, let us recall, is given by the formula

$$\begin{aligned} \partial f(g_1, \dots, g_{n+1}) &= g_1 f(g_2, \dots, g_{n+1}) + \sum_{i=1}^n (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &\quad + (-1)^{n+1} f(g_1, \dots, g_n), \end{aligned}$$

is functorial in S and therefore defines a homomorphism

$$\partial: C^n(\mathbf{G}, \mathbf{F}) \longrightarrow C^{n+1}(\mathbf{G}, \mathbf{F})$$

such that $\partial \circ \partial = 0$. One has thus defined a complex of abelian groups (and even of $\Gamma(\mathbf{O})$ -modules), denoted $C^*(\mathbf{G}, \mathbf{F})$. One defines in the same way the complex of $\mathbf{O}$ -modules $\underline{C}^*(\mathbf{G}, \mathbf{F})$, and one has

$$C^*(\mathbf{G}, \mathbf{F}) = \Gamma(\underline{C}^*(\mathbf{G}, \mathbf{F})).$$

One denotes by $H^n(\mathbf{G}, \mathbf{F})$ (resp. $\underline{H}^n(\mathbf{G}, \mathbf{F})$) the homology groups (resp. the homology $\widehat{\mathcal{C}}$ -groups) of the complex $C^*(\mathbf{G}, \mathbf{F})$ (resp. $\underline{C}^*(\mathbf{G}, \mathbf{F})$).

One has in particular

$$\underline{H}^0(\mathbf{G}, \mathbf{F}) = \mathbf{F}^{\mathbf{G}} \quad \text{and} \quad H^0(\mathbf{G}, \mathbf{F}) = \Gamma(\mathbf{F}^{\mathbf{G}}).$$

Remark 5.1.1. — ⁽⁴⁵⁾The “set-theoretic” description of ∂ given above is convenient for checking that $\partial \circ \partial = 0$. One can also define ∂ in terms of the multiplication $m: \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}$ and of the action $\mu: \mathbf{G} \times \mathbf{F} \longrightarrow \mathbf{F}$ as follows: for every $f \in C^n(\mathbf{G}, \mathbf{F})$,

$$\partial f = \mu \circ (\text{id}_{\mathbf{G}} \times f) + \sum_{i=1}^n (-1)^i f \circ (\text{id}_{\mathbf{G}^{i-1}} \times m \times \text{id}_{\mathbf{G}^{n-i}}) + (-1)^{n+1} f \circ \text{pr}_{[1,n]},$$

where $\text{pr}_{[1,n]}$ denotes the projection of $\mathbf{G}^{n+1} = \mathbf{G}^n \times \mathbf{G}$ onto $\mathbf{G}^n$. Likewise, for every $S \in \text{Ob}(\mathcal{C})$ and $f \in \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S)$, one has

$$\partial f = \mu_S \circ (\text{id}_{\mathbf{G}_S} \times f) + \sum_{i=1}^n (-1)^i f \circ (\text{id}_{\mathbf{G}_S^{i-1}} \times m_S \times \text{id}_{\mathbf{G}_S^{n-i}}) + (-1)^{n+1} f \circ \text{pr}_{[1,n]},$$

where m_S and μ_S are deduced from m and μ by base change.

5.2. — ⁽⁴⁶⁾One recalls (cf. § 3) that $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ is endowed with the structure of an abelian category, defined “argument by argument”; thus,

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

is an exact sequence of $\mathbf{G}$ - $\mathbf{O}$ -modules if, and only if, the sequence of abelian groups

$$0 \longrightarrow \mathbf{F}'(S) \longrightarrow \mathbf{F}(S) \longrightarrow \mathbf{F}''(S) \longrightarrow 0$$

is exact, for every $S \in \text{Ob}(\mathcal{C})$.

⁽⁴⁴⁾**N.D.E.** This complex is often called the “Hochschild complex”; see for example § II.3 of [DG70].

⁽⁴⁵⁾**N.D.E.** This remark has been added.

⁽⁴⁶⁾**N.D.E.** The following reminder has been added.

⁽⁴⁷⁾ Suppose $\mathcal{C}$ is small; then, by 3.2.1, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects, so that the derived functors of the left exact functors H^0 and $\underline{H}^0$ are defined. We now propose to show that the functors H^n (resp. $\underline{H}^n$) are indeed the derived functors of H^0 (resp. $\underline{H}^0$).

Definition 5.2.0. — ⁽⁴⁸⁾For every $\mathbf{O}$ -module $\mathbf{P}$, one denotes by $E(\mathbf{P})$ the object $\underline{\text{Hom}}(\mathbf{G}, \mathbf{P})$ of $\widehat{\mathcal{C}}$ endowed with the structure of $\mathbf{G}\text{-}\mathbf{O}$ -module defined as follows: for every $S \in \text{Ob}(\mathcal{C})$ one has $\underline{\text{Hom}}(\mathbf{G}, \mathbf{P})(S) = \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$, and one lets $g \in \mathbf{G}(S)$ and $a \in \mathbf{O}(S)$ operate on $\phi \in \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ by the formulas

$$(g \cdot \phi)(h) = \phi(hg) \quad \text{and} \quad (a \cdot \phi)(h) = a\phi(h),$$

for every $h \in \mathbf{G}(S')$, $S' \rightarrow S$. Moreover, for every $\phi \in \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ one sets

$$\varepsilon(\phi) = \phi(1) \in \mathbf{P}(S)$$

(where 1 denotes the unit element of $\mathbf{G}(S)$).

This defines a functor $E: (\mathbf{O}\text{-Mod.}) \rightarrow (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ and a natural transformation $\varepsilon: E \rightarrow \text{Id}$, where Id denotes the identity functor of $(\mathbf{O}\text{-Mod.})$.

Remark 5.2.0.1. — ⁽⁴⁸⁾In what follows, let us denote by $\mathbf{G}_1$ and $\mathbf{G}_2$ two copies of $\mathbf{G}$. Then the morphism

$$\mathbf{G}_1 \times E(\mathbf{P}) \rightarrow E(\mathbf{P}), \quad (g_1, \phi) \mapsto (g_2 \mapsto \phi(g_2 g_1))$$

corresponds, via the isomorphisms

$$\begin{aligned} \text{Hom}(\mathbf{G}_1 \times E(\mathbf{P}), E(\mathbf{P})) &\simeq \text{Hom}(E(\mathbf{P}), \underline{\text{Hom}}(\mathbf{G}_1, \underline{\text{Hom}}(\mathbf{G}_2, \mathbf{P}))) \\ &\simeq \text{Hom}(E(\mathbf{P}), \underline{\text{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})) \end{aligned}$$

to the morphism $\phi \mapsto ((g_2, g_1) \mapsto \phi(g_2 g_1))$, i.e. to the morphism

$$\underline{\text{Hom}}(\mathbf{G}, \mathbf{P}) \rightarrow \underline{\text{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})$$

induced by the multiplication $\mu_{\mathbf{G}}: \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$, $(g_2, g_1) \mapsto g_2 g_1$.

Lemma 5.2.0.2. — ⁽⁴⁸⁾

- (i) *The functor E is right adjoint to the forgetful functor $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}) \rightarrow (\mathbf{O}\text{-Mod.})$; more precisely, $\varepsilon: E \rightarrow \text{Id}$ induces, for every*

$$\mathbf{M} \in (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}) \quad \text{and} \quad \mathbf{P} \in \text{Ob}(\mathbf{O}\text{-Mod.}),$$

a bijection

$$\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}\text{-Mod.}}(\mathbf{M}, \mathbf{P})$$

functorial in $\mathbf{M}$ and $\mathbf{P}$.

- (ii) *Consequently, if $\mathbf{I}$ is an injective object of $(\mathbf{O}\text{-Mod.})$ then $E(\mathbf{I})$ is an injective object of $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$.*

PROOF. To every $\mathbf{O}$ -morphism $f: \mathbf{M} \rightarrow \mathbf{P}$, one associates the element ϕ_f of

$$\text{Hom}_{\mathbf{O}}(\mathbf{M}, E(\mathbf{P}))$$

defined as follows. For every $S \in \text{Ob}(\mathcal{C})$ and $m \in \mathbf{M}(S)$, $\phi_f(m)$ is the element of $\text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ defined by: for every $g \in \mathbf{G}(S')$, $S' \rightarrow S$,

$$\phi_f(m)(g) = f(gm) \in \mathbf{P}(S').$$

Then, for every $h \in \mathbf{G}(S)$, one has $\phi_f(hm) = h \cdot f(m)$, i.e. ϕ_f is an element of

$$\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})).$$

⁽⁴⁷⁾**N.D.E.** The following sentence has been added.

⁽⁴⁸⁾**N.D.E.** The original has been modified, in order to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

If

$$\phi \in \text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P}))$$

and if one denotes, for every $m \in \mathbf{M}(S)$, $f(m) = \phi(m)(1)$, then

$$\phi_f(m)(g) = f(gm) = \phi(gm)(1) = (g \cdot \phi(m))(1) = \phi(m)(g),$$

i.e. $\phi_f = \phi$. Conversely, it is clear that $\phi_f(m)(1) = f(m)$. This proves (i), and (ii) follows from it at once. $\square$

Definition 5.2.0.3. — Let $\mathbf{M}$ be a $\mathbf{G}\text{-}\mathbf{O}$ -module; the identity map of $\mathbf{M}$ (considered as an $\mathbf{O}$ -module) corresponds by adjunction to the morphism of $\mathbf{G}\text{-}\mathbf{O}$ -modules

$$\mu_{\mathbf{M}}: \mathbf{M} \longrightarrow E(\mathbf{M})$$

such that for every $S \in \text{Ob}(\mathcal{C})$ and $m \in \mathbf{M}(S)$, $\mu_{\mathbf{M}}(m)$ is the morphism $\mathbf{G}_S \longrightarrow \mathbf{M}_S$ defined by: for every $S' \longrightarrow S$ and $g \in \mathbf{G}(S')$, $\mu_{\mathbf{M}}(m)(g) = g \cdot m_{S'} \in \mathbf{M}(S')$.

Note that $\mu_{\mathbf{M}}$ is a *monomorphism*. Indeed, $\varepsilon_{\mathbf{M}}: E(\mathbf{M}) \longrightarrow \mathbf{M}$ is a morphism of $\mathbf{O}$ -modules such that $\varepsilon_{\mathbf{M}} \circ \mu_{\mathbf{M}} = \text{id}_{\mathbf{M}}$; consequently $\mathbf{M}$ is a direct factor, as an $\mathbf{O}$ -module, of $E(\mathbf{M})$.

Proposition 5.2.1. — *One assumes that $\mathcal{C}$ is small, that finite products exist in it, and that $\mathbf{G}$ is representable. Then the functors $H^n(\mathbf{G}, _)$ (resp. $\underline{H}^n(\mathbf{G}, _)$) are the derived functors of the left exact functor $H^0(\mathbf{G}, _)$ (resp. $\underline{H}^0(\mathbf{G}, _)$) on the category of $\mathbf{G}\text{-}\mathbf{O}$ -modules.*

By virtue of the well-known general results⁽⁴⁹⁾, it suffices to check that the $H^n(\mathbf{G}, _)$ (resp. $\underline{H}^n(\mathbf{G}, _)$) form an effaceable cohomological functor in degrees > 0 .

Let

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

be an exact sequence of $\mathbf{G}\text{-}\mathbf{O}$ -modules, and let $S \in \text{Ob}(\mathcal{C})$. By hypothesis, $\mathbf{G}$ is representable by an object $G \in \text{Ob}(\mathcal{C})$, and finite products exist in $\mathcal{C}$; in particular $\mathcal{C}$ has a final element e . Thus each $G^n \times \mathbf{h}_S$ is representable by $G^n \times S$ (with $G^0 = e$), and the sequence

$$0 \longrightarrow \mathbf{F}'(G^n \times S) \longrightarrow \mathbf{F}(G^n \times S) \longrightarrow \mathbf{F}''(G^n \times S) \longrightarrow 0$$

is exact. This shows that the sequence of $\mathbf{O}$ -modules

$$0 \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}') \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}) \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}'') \longrightarrow 0$$

is exact. It follows that $\underline{C}^*(\mathbf{G}, _)$, considered as a functor on $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ with values in the category of complexes of $(\mathbf{O}\text{-Mod.})$, is exact. This shows that the $\underline{H}^n(\mathbf{G}, _)$ do indeed form a cohomological functor. As the functor Γ is exact, the same holds for the $H^n(\mathbf{G}, _)$. It will now suffice for us to prove:

Lemma 5.2.2. *For every $\mathbf{P} \in \text{Ob}(\mathbf{O}\text{-Mod.})$, one has:*

$$H^n(\mathbf{G}, \text{Hom}(\mathbf{G}, \mathbf{P})) = 0 \quad \text{and} \quad \underline{H}^n(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \quad \text{for } n > 0.$$

It suffices for us to prove that $C^*(\mathbf{G}, \text{Hom}(\mathbf{G}, \mathbf{P}))$ and $\underline{C}^*(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$ are homotopically trivial in degrees > 0 . It even suffices to do so for the second, the corresponding result for the first being deduced from it by base change.⁽⁵⁰⁾ Now, one defines for every $n \geq 0$ a morphism

$$\sigma: \underline{C}^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P})) \longrightarrow \underline{C}^n(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$$

as follows. Let $f \in \underline{C}^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$; for every $S \in \text{Ob}(\mathcal{C})$ and $g_1, \dots, g_n \in \mathbf{G}(S)$, $\sigma(f)(g_1, \dots, g_n)$ is the element of $\text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ defined by: for every $S' \longrightarrow S$ and $x \in \mathbf{G}(S')$,

$$\sigma(f)(g_1, \dots, g_n)(x) = f(x, g_1, \dots, g_n)(e) \in \mathbf{P}(S')$$

⁽⁴⁹⁾**N.D.E.** Cf. [Gr57], 2.2.1 and 2.3. Moreover, the original has been spelled out in what follows.

⁽⁵⁰⁾**N.D.E.** The original has been expanded in what follows.

(where e denotes the unit element of $\mathbf{G}(S')$). Then σ is a homotopy operator in degrees > 0 . Indeed, for every $g_1, \dots, g_{n+1} \in \mathbf{G}(S)$ and $x \in \mathbf{G}(S')$ one has, on the one hand:

$$\begin{aligned} \partial\sigma(f)(g_1, \dots, g_{n+1})(x) &= f(xg_1, g_2, \dots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^n (-1)^i f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1})(e) \\ &\quad + (-1)^{n+1} f(x, g_1, \dots, g_n)(e), \end{aligned}$$

and on the other hand:

$$\begin{aligned} \sigma(\partial f)(g_1, \dots, g_{n+1})(x) &= (xf(g_1, g_2, \dots, g_{n+1}))(e) - f(xg_1, g_2, \dots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^n (-1)^{i+1} f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &\quad + (-1)^{n+2} f(x, g_1, \dots, g_n)(e), \end{aligned}$$

whence

$$(\partial\sigma(f) + \sigma(\partial f))(g_1, \dots, g_{n+1})(x) = f(g_1, \dots, g_{n+1})(x),$$

i.e. $\partial\sigma + \sigma\partial$ is the identity map of $\underline{C}^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$, for every $n \geq 0$.

Remark 5.2.3. ⁽⁵¹⁾ The hypothesis “ $\mathcal{C}$ small” is used only to ensure the existence of the derived functors $R^n H^0$ and $R^n \underline{H}^0$. In all cases, what precedes shows that the functors $H^n(\mathbf{G}, \)$ resp. $\underline{H}^n(\mathbf{G}, \)$ form a cohomological functor, effaceable in degrees > 0 , hence they are the (right) satellite functors of the left exact functor $H^0(\mathbf{G}, \)$ resp. $\underline{H}^0(\mathbf{G}, \)$, in the sense of [Gr57], 2.2; if $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects (which is the case if $\mathcal{C}$ is small), they coincide with the derived functors (*loc. cit.* 2.3).

5.3. Cohomology of the $G\text{-}\mathcal{O}_S$ -modules. — Let S be a scheme, G an S -group and $\mathcal{F}$ a quasi-coherent $G\text{-}\mathcal{O}_S$ -module. One defines the cohomology groups of G with values in $\mathcal{F}$ by

$$H^n(G, \mathcal{F}) = H^n(\mathbf{h}_G, \mathbf{W}(\mathcal{F})).$$

(for the notations, cf. 4.6).

Suppose G is *affine* over S . Then, in view of proposition 4.6.4, this cohomology is computed in the following way: $H^n(G, \mathcal{F})$ is the n -th homology group of the complex $C^*(G, \mathcal{F})$ whose n -th term is:

$$C^n(G, \mathcal{F}) = \Gamma\left(S, \mathcal{F} \otimes \underbrace{\mathcal{A}(G) \otimes \dots \otimes \mathcal{A}(G)}_{n \text{ times}}\right).$$

If f resp. a_i is a section of $\mathcal{F}$ resp. of $\mathcal{A}(G)$ over an open of S , one has

$$\begin{aligned} \partial(f \otimes a_1 \otimes \dots \otimes a_n) &= \mu_{\mathcal{F}}(f) \otimes a_1 \otimes a_2 \otimes \dots \otimes a_n \\ &\quad + \sum_{i=1}^n (-1)^i f \otimes a_1 \otimes \dots \otimes \Delta a_i \otimes \dots \otimes a_n \\ &\quad + (-1)^{n+1} f \otimes a_1 \otimes a_2 \otimes \dots \otimes a_n \otimes 1, \end{aligned}$$

where $\Delta: \mathcal{A}(G) \rightarrow \mathcal{A}(G) \otimes \mathcal{A}(G)$ and $\mu_{\mathcal{F}}: \mathcal{F} \rightarrow \mathcal{F} \otimes \mathcal{A}(G)$ describe the coalgebra structure of $\mathcal{A}(G)$ and the comodule structure of $\mathcal{F}$. Let us note in passing that the cohomology of G with values in $\mathcal{F}$ therefore depends only on the comodule structure of $\mathcal{F}$, and in particular only on the S -monoid structure of G .

One has in particular

$$H^0(G, \mathcal{F}) = \Gamma(S, \mathcal{F}^G),$$

⁽⁵¹⁾**N.D.E.** This remark has been added.

where $\mathcal{F}^G$, the *sheaf of invariants* of $\mathcal{F}$, is defined as the sheaf whose sections over the open U of S are the sections of $\mathcal{F}$ over U whose inverse image in every S' over U is invariant under $G(S')$ (cf. 4.7.1.2).

Theorem 5.3.1. *Let S be an affine scheme, G an S -group affine and flat over S . The functors $H^n(G, _)$ are the derived functors of $H^0(G, _)$ on the category of quasi-coherent G - $\mathcal{O}_S$ -modules.*

PROOF. ⁽⁵²⁾ As G is affine and flat over S , then, by 4.7.2.1, the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ is equivalent to the category $(\mathcal{A}(G)\text{-Comod.q.c.})$ of quasi-coherent $\mathcal{A}(G)$ -comodules on $\mathcal{O}_S$, and is therefore abelian. On the other hand, $\mathcal{A}(G)$ being a flat $\mathcal{O}_S$ -module, each functor $\mathcal{F} \mapsto \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G)^{\otimes n}$ is exact; as moreover S is affine, one obtains that $C^*(G, _)$ is an exact functor on $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

Let us denote by Δ resp. η the comultiplication resp. the augmentation of $\mathcal{A}(G)$. For every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{P}$, one denotes $\text{Ind}(\mathcal{P}) = \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ equipped with the $\mathcal{A}(G)$ -comodule structure defined by

$$\text{id}_{\mathcal{P}} \otimes \Delta: \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G);$$

this defines a functor $\text{Ind}: (\mathcal{O}_S\text{-Mod.q.c.}) \longrightarrow (G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

It follows from 4.6.4.1, 5.2.0 and 5.2.0.1 that one has an isomorphism of G - $\mathcal{O}_S$ -modules:

$$(*) \quad \mathbf{W}(\text{Ind}(\mathcal{P})) \simeq E(\mathbf{W}(\mathcal{P})) = \text{Hom}(G, \mathbf{W}(\mathcal{P})).$$

Via this identification, the morphism $\varepsilon: E(\mathbf{W}(\mathcal{P})) \longrightarrow \mathbf{W}(\mathcal{P})$ corresponds to the morphism of $\mathcal{O}_S$ -modules $\text{id}_{\mathcal{P}} \otimes \eta: \text{Ind}(\mathcal{P}) \longrightarrow \mathcal{P}$.

One has already used that the functor $\mathbf{W}: (\mathcal{O}_S\text{-Mod.}) \longrightarrow (\mathcal{O}_S\text{-Mod.})$ is fully faithful; the same holds, by definition 4.7.1, of its restriction to $(G\text{-}\mathcal{O}_S\text{-Mod.})$, i.e. if $\mathcal{M}, \mathcal{M}'$ are G - $\mathcal{O}_S$ -modules, one has a functorial isomorphism

$$\text{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathcal{M}, \mathcal{M}') \simeq \text{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathbf{W}(\mathcal{M}), \mathbf{W}(\mathcal{M}')).$$

Consequently, one deduces from lemma 5.2.0.2 the

Corollary 5.3.1.1. *(i) The functor Ind is right adjoint to the forgetful functor*

$$(G\text{-}\mathcal{O}_S\text{-Mod.q.c.}) \longrightarrow (\mathcal{O}_S\text{-Mod.q.c.}).$$

More precisely, the map

$$\text{id}_{\mathcal{P}} \otimes \eta: \text{Ind}(\mathcal{P}) \longrightarrow \mathcal{P}$$

induces, for every object $\mathcal{M}$ of $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$, a bijection

$$\text{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathcal{M}, \text{Ind}(\mathcal{P})) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_S}(\mathcal{M}, \mathcal{P}).$$

(ii) Consequently, if $\mathcal{I}$ is an injective object of $(\mathcal{O}_S\text{-Mod.q.c.})$ then $\text{Ind}(\mathcal{I})$ is an injective object of $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

Let $\mathcal{F}$ be a G - $\mathcal{O}_S$ -module and $\mu_{\mathcal{F}}: \mathcal{F} \longrightarrow \text{Ind}(\mathcal{F})$ the map defining the $\mathcal{A}(G)$ -comodule structure. It follows from 5.2.0.3 (or else from the axioms (CM 1) and (CM 2) of 4.7.2) that $\mu_{\mathcal{F}}$ is a morphism of G - $\mathcal{O}_S$ -modules, and that $(\text{id}_{\mathcal{F}} \otimes \eta) \circ \mu_{\mathcal{F}} = \text{id}_{\mathcal{F}}$, hence that $\mathcal{F}$ is a direct factor of $\text{Ind}(\mathcal{F})$ as an $\mathcal{O}_S$ -module; in particular, $\mu_{\mathcal{F}}$ is a monomorphism. As one has, by (*) and 5.2.2,

$$H^n(G, \mathbf{W}(\text{Ind}(\mathcal{F}))) \simeq H^n(G, \text{Hom}_S(G, \mathbf{W}(\mathcal{F}))) = 0 \quad \text{for } n > 0,$$

one therefore obtains that $H^n(G, _)$ is effaceable for $n > 0$.

Finally, S being affine, $(\mathcal{O}_S\text{-Mod.q.c.})$ has enough injective objects. Let therefore $\mathcal{F} \hookrightarrow \mathcal{I}$ be a monomorphism of $\mathcal{O}_S$ -modules, where $\mathcal{I}$ is an injective object of $(\mathcal{O}_S\text{-Mod.q.c.})$; then, $\mathcal{A}(G)$ being flat over $\mathcal{O}_S$, $\text{Ind}(\mathcal{F})$ is a sub- G - $\mathcal{O}_S$ -module of $\text{Ind}(\mathcal{I})$, whence:

⁽⁵²⁾**N.D.E.** The original has been modified, in order to show that the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ is abelian and has enough injective objects. One may compare with [Ja03], Part I, 3.3–3.4, 3.9, 4.2 and 4.14–4.16 (where one will take care that “ k -group scheme” means “affine k -group scheme”, cf. *loc. cit.*, 2.1).

Corollary 5.3.1.2. *The abelian category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ has enough injective objects.*

Taking into account [Gr57], 2.2.1 and 2.3 (already used in the proof of 5.2.1), this completes the proof of theorem 5.3.1. $\square$

Remark 5.3.1.3. One can also prove 5.3.1.1 by the following computation. To every morphism of $G\text{-}\mathcal{O}_S$ -modules $\phi: \mathcal{M} \rightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ one associates the $\mathcal{O}_S$ -morphism $(\text{id}_{\mathcal{P}} \otimes \eta) \circ \phi: \mathcal{M} \rightarrow \mathcal{P}$. Conversely, to every $\mathcal{O}_S$ -morphism $f: \mathcal{M} \rightarrow \mathcal{P}$ one associates the morphism of $G\text{-}\mathcal{O}_S$ -modules $(f \otimes \text{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}}: \mathcal{M} \rightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$. One sees at once that

$$(\text{id}_{\mathcal{P}} \otimes \eta) \circ (f \otimes \text{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}} = (f \otimes \text{id}_{\mathcal{O}_S}) \circ (\text{id}_{\mathcal{P}} \otimes \eta) \circ \mu_{\mathcal{M}} = f.$$

On the other hand, for every ϕ the diagram below is commutative:

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{\phi} & \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ \mu_{\mathcal{M}} \downarrow & & \downarrow \text{id}_{\mathcal{P}} \otimes \Delta \\ \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{A}(G) & \xrightarrow[\phi \otimes \text{id}_{\mathcal{A}(G)}]{} & \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G). \end{array}$$

It follows that

$$\begin{aligned} ((\text{id}_{\mathcal{P}} \otimes \eta) \circ \phi) \otimes \text{id}_{\mathcal{A}(G)} \circ \mu_{\mathcal{M}} &= (\text{id}_{\mathcal{P}} \otimes \eta \otimes \text{id}_{\mathcal{A}(G)}) \circ (\phi \otimes \text{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}} \\ &= (\text{id}_{\mathcal{P}} \otimes \eta \otimes \text{id}_{\mathcal{A}(G)}) \circ (\text{id}_{\mathcal{P}} \otimes \Delta) \circ \phi = \phi. \end{aligned}$$

This proves 5.3.1.1 (i) (and (ii) follows).

Let $\mathcal{F}$ be a $G\text{-}\mathcal{O}_S$ -module; one has seen above that axiom (CM 2) of 4.7.2 shows that, considered as an $\mathcal{O}_S$ -module, $\mathcal{F}$ is a direct factor of $E(\mathcal{F})$. This entails:

Proposition 5.3.2. *Let S be an affine scheme and G an affine and flat S -group; suppose that every exact sequence $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$ of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules, which splits as a sequence of $\mathcal{O}_S$ -modules, likewise splits as a sequence of $G\text{-}\mathcal{O}_S$ -modules.*

Then the functors $H^n(G, _)$ are zero for $n > 0$ (or, what amounts to the same, the functor $H^0(G, _)$ is exact).

Indeed, by the hypothesis, the sequence of $G\text{-}\mathcal{O}_S$ -modules

$$0 \rightarrow \mathcal{F} \rightarrow E(\mathcal{F}) \rightarrow E(\mathcal{F})/\mathcal{F} \rightarrow 0$$

is split; $\mathcal{F}$ is therefore a direct factor, as a $G\text{-}\mathcal{O}_S$ -module, in $E(\mathcal{F})$, but the cohomology of the latter is zero.

One deduces immediately from this and from proposition 4.7.4:

Theorem 5.3.3. *Let S be an affine scheme and G a diagonalizable S -group. For every quasi-coherent $G\text{-}\mathcal{O}_S$ -module $\mathcal{F}$, one has $H^n(G, \mathcal{F}) = 0$, for $n > 0$.*

Remark. Proposition 5.3.2 remains valid when G is not necessarily flat over S ; the proof then appeals to relative cohomology.⁽⁵³⁾

6. G -equivariant objects and modules

⁽⁵⁴⁾ Let $\mathcal{C}$ be a category having a final object and in which fiber products exist. Let $\mathbf{G}$ be a $\hat{\mathcal{C}}$ -group, $\pi: \mathbf{M} \rightarrow \mathbf{X}$ a morphism of $\hat{\mathcal{C}}$, and $\lambda = \lambda_{\mathbf{X}}: \mathbf{G} \times \mathbf{X} \rightarrow \mathbf{X}$ an action of $\mathbf{G}$ on $\mathbf{X}$. In what follows, one will denote by $\mathbf{Y} \times_f \mathbf{M}$ the fiber product of $\pi: \mathbf{M} \rightarrow \mathbf{X}$ and of an $\mathbf{X}$ -functor $f: \mathbf{Y} \rightarrow \mathbf{X}$.

⁽⁵³⁾**N.D.E.** The editors have not sought to develop this remark.

⁽⁵⁴⁾**N.D.E.** This section has been added.

For every $U \in \text{Ob}(\mathcal{C})$ and $x \in \mathbf{X}(U)$, one will denote $\mathbf{M}_x = U \times_x \mathbf{M}$, i.e. for every $\varphi: U' \rightarrow U$, one has

$$\mathbf{M}_x(U') = \{m \in \mathbf{M}(U') \mid \pi(m) = x_{U'} = \varphi^*(x)\}.$$

Finally, if $g \in \mathbf{G}(U)$ one will also denote by gx the element $\lambda(g, x)$ of $\mathbf{X}(U)$.

Definition 6.1. a) One says that $\mathbf{M}$ is a *G-equivariant X-object* if one has given oneself an action $\Lambda: \mathbf{G} \times \mathbf{M} \rightarrow \mathbf{M}$ of $\mathbf{G}$ on $\mathbf{M}$ lifting λ , i.e. such that the square below is commutative:

$$\begin{array}{ccc} \mathbf{G} \times \mathbf{M} & \xrightarrow{\Lambda} & \mathbf{M} \\ \text{id}_{\mathbf{G}} \times \pi \downarrow & & \downarrow \pi \\ \mathbf{G} \times \mathbf{X} & \xrightarrow{\lambda} & \mathbf{X}. \end{array}$$

This amounts to giving, for every morphism $(g, x): U \rightarrow \mathbf{G} \times \mathbf{X}$, maps

$$\Lambda_x^U(g): \mathbf{M}_x(U) \rightarrow \mathbf{M}_{gx}(U), \quad m \mapsto g \cdot m$$

satisfying $1 \cdot m = m$ and $g \cdot (h \cdot m) = (gh) \cdot m$ and functorial in the $(\mathbf{G} \times \mathbf{X})$ -object U . This amounts again to giving morphisms of U -objects:

$$\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$$

satisfying $\Lambda_x(1) = \text{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$.

b) Let now $\mathbf{O}$ be a $\widehat{\mathcal{C}}$ -ring and let $\mathbf{O}_{\mathbf{X}} = \mathbf{O} \times \mathbf{X}$. Under the conditions of (a), one says that $\mathbf{M}$ is a *G-equivariant O_X-module* if it is an $\mathbf{O}_{\mathbf{X}}$ -module (cf. definition 4.3.3.1, valid for every ring-valued functor on a category $\mathcal{C}$) and if the action Λ is compatible with the $\mathbf{O}_{\mathbf{X}}$ -module structure of $\mathbf{M}$, i.e. if for every $(g, x) \in \mathbf{G}(U) \times \mathbf{X}(U)$ ($U \in \text{Ob}(\mathcal{C})$), the map $\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$, $m \mapsto g \cdot m$ is a morphism of $\mathbf{O}_U$ -modules.

Remark 6.2. (i) In (a) above, the conditions $\Lambda_x(1) = \text{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ evidently entail that each $\Lambda_x(g)$ is an *isomorphism*, with inverse $\Lambda_{gx}(g^{-1})$. Conversely, if one supposes that each $\Lambda_x(g)$ is an isomorphism, the condition $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ applied to $h = 1$ gives $\Lambda_x(1) = \text{id}$.

(ii) Let $\mathbf{M}$ be an $\mathbf{O}_{\mathbf{X}}$ -module. First, one sees that giving a morphism $\Lambda: \mathbf{G} \times \mathbf{M} \rightarrow \mathbf{M}$ making the diagram of 6.1 commutative and such that each morphism $\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$, $m \mapsto g \cdot m$, is an isomorphism of $\mathbf{O}_U$ -modules, amounts to giving an *isomorphism* of $\mathbf{O}_{\mathbf{G} \times \mathbf{X}}$ -modules:

$$\begin{aligned} \theta: \mathbf{G} \times \mathbf{M} &= (\mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}}} \mathbf{M} \xrightarrow{\sim} (\mathbf{G} \times \mathbf{X}) \times_{\Lambda} \mathbf{M} \\ (g, x, m) &\mapsto (g, x, g \cdot m). \end{aligned}$$

As one has supposed that each $\Lambda_x(g)$ is an isomorphism, the equality $\Lambda_x(1) = \text{id}$ will be a consequence of the equality $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ applied to $h = 1$. One therefore obtains that Λ is an *action* of $\mathbf{G}$ on $\mathbf{M}$ (i.e. $g \cdot (h \cdot m) = (gh) \cdot m$) if and only if the diagram of $(\mathbf{G} \times \mathbf{G} \times \mathbf{X})$ -isomorphisms below is commutative (where one denotes by μ the multiplication of $\mathbf{G}$ and by $f^*(\theta)$ the isomorphism deduced from θ by a base change $f: \mathbf{G} \times \mathbf{G} \times \mathbf{X} \rightarrow \mathbf{G} \times \mathbf{X}$):

$$\begin{array}{ccc} (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ \text{pr}_{2,3}} \mathbf{M} & \xrightarrow[\sim]{\text{pr}_{2,3}^*(\theta)} & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ \text{pr}_{2,3}} \mathbf{M} \\ \\ (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ (\mu \times \text{id}_{\mathbf{X}})} \mathbf{M} & & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ (\text{id}_{\mathbf{G}} \times \lambda)} \mathbf{M} \\ (\mu \times \text{id}_{\mathbf{X}})^*(\theta) \downarrow \sim & & (\text{id}_{\mathbf{G}} \times \lambda)^*(\theta) \downarrow \sim \\ (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ (\mu \times \text{id}_{\mathbf{X}})} \mathbf{M} & & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ (\text{id}_{\mathbf{G}} \times \lambda)} \mathbf{M}. \end{array}$$

One therefore sees that giving a structure of G -equivariant $\mathbf{O}_X$ -module on $\mathbf{M}$ amounts to giving an isomorphism θ of $\mathbf{O}_{G \times X}$ -modules as above, such that the diagram above is commutative.

(iii) Everything that precedes extends to the case where $\mathbf{G}$ is only a $\widehat{\mathcal{C}}$ -monoid: in this case, giving an action $\Lambda: \mathbf{G} \times \mathbf{M} \rightarrow \mathbf{M}$ lifting λ and such that each $\Lambda_x(g): \mathbf{M}_x \rightarrow \mathbf{M}_{gx}$ is a morphism of $\mathbf{O}_U$ -modules, amounts to giving a *morphism* θ of $\mathbf{O}_{G \times X}$ -modules as in (ii), such that the diagram above (without the signs $\sim$ under the arrows) is commutative, and such that $p_{\mathbf{M}} \circ \theta \circ (\varepsilon_{\mathbf{G}} \times \text{id}_{\mathbf{M}}) = \text{id}_{\mathbf{M}}$, where $\varepsilon_{\mathbf{G}}$ denotes the unit section of $\mathbf{G}$ and $p_{\mathbf{M}}$ the projection onto $\mathbf{M}$.

6.3. G -equivariant morphisms. — Let $\mathbf{Y}$ be a second object of $\widehat{\mathcal{C}}$, equipped with an action $\lambda_{\mathbf{Y}}: \mathbf{G} \times \mathbf{Y} \rightarrow \mathbf{Y}$ of $\mathbf{G}$, and let $\mathbf{N}$ be a second G -equivariant $\mathbf{O}_X$ -module. One says that a $\widehat{\mathcal{C}}$ -morphism $f: \mathbf{Y} \rightarrow \mathbf{X}$ resp. a morphism of $\mathbf{O}_X$ -modules $\phi: \mathbf{M} \rightarrow \mathbf{N}$ is *G -equivariant* if it commutes with the action of $\mathbf{G}$, i.e. if one has, set-theoretically, $f(g \cdot y) = g \cdot f(y)$ resp. $\phi(g \cdot m) = g \cdot \phi(m)$, which amounts to saying that $f \circ \lambda_{\mathbf{Y}} = \lambda_{\mathbf{X}} \circ (\text{id}_{\mathbf{G}} \times f)$ resp. $\phi \circ \Lambda_{\mathbf{M}} = \Lambda_{\mathbf{N}} \circ (\text{id}_{\mathbf{G}} \times \phi)$.

One then obtains at once the following lemma:

Lemma 6.3.1. *Let $f: \mathbf{Y} \rightarrow \mathbf{X}$ be a G -equivariant morphism and $\mathbf{M}$ a G -equivariant $\mathbf{O}_X$ -module. Then the inverse image $f^*(\mathbf{M}) = \mathbf{Y} \times_f \mathbf{M}$ is a G -equivariant $\mathbf{O}_Y$ -module.*

On the other hand, $\mathbf{G}$ acts on $\text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{Y}, \mathbf{X})$. Indeed, let $T \in \text{Ob}(\mathcal{C})$, $g \in \mathbf{G}(T)$ and φ a T -morphism $\mathbf{Y}_T \rightarrow \mathbf{X}_T$; then g defines automorphisms $\lambda_{\mathbf{Y}}(g)$ and $\lambda_{\mathbf{X}}(g)$ of $\mathbf{Y}_T$ and $\mathbf{X}_T$, and one will denote by $g \cdot \varphi$ (or also $g\varphi g^{-1}$) the morphism $\lambda_{\mathbf{X}}(g) \circ \varphi \circ \lambda_{\mathbf{Y}}(g^{-1})$. This defines an action of $\mathbf{G}(T)$ on $\text{Hom}_T(\mathbf{Y}_T, \mathbf{X}_T)$, functorial in T .

Definition 6.3.2. If $\varphi: \mathbf{Y} \rightarrow \mathbf{X}$ is an arbitrary $\widehat{\mathcal{C}}$ -morphism, one may therefore consider its stabilizer $\text{Stab}_{\mathbf{G}}(\varphi)$ (cf. 2.3.3.1): for every $T \in \text{Ob}(\mathcal{C})$, $\text{Stab}_{\mathbf{G}}(\varphi)(T)$ is the subgroup of $\mathbf{G}(T)$ formed of the g such that $g \circ \varphi_T = \varphi_T \circ g$, i.e. such that the diagram

$$\begin{array}{ccc} \mathbf{Y}_T & \xrightarrow{\varphi_T} & \mathbf{X}_T \\ g \downarrow & & \downarrow g \\ \mathbf{Y}_T & \xrightarrow{\varphi_T} & \mathbf{X}_T \end{array}$$

commutes. Then the morphism $\varphi: \mathbf{Y} \rightarrow \mathbf{X}$ is equivariant under $\text{Stab}_{\mathbf{G}}(\varphi)$.

6.4. Global sections. — Let $\mathbf{M}$ be a G -equivariant $\mathbf{O}_X$ -module. Denote by S_0 the final object of $\mathcal{C}$ and (cf. Exp. II, 1.1) by $\prod_{\mathbf{X}/S_0} \mathbf{M}$ the “functor of sections of $\mathbf{M}$ over $\mathbf{X}$ ”: it is the functor which to every $T \in \text{Ob}(\mathcal{C})$ associates

$$\text{Hom}_{\mathbf{X}}(\mathbf{X}_T, \mathbf{M}) = \text{Hom}_{\mathbf{X}_T}(\mathbf{X}_T, \mathbf{M}_T) = \Gamma(\mathbf{M}_T/\mathbf{X}_T).$$

Let us recall on the other hand that every morphism $g: \mathbf{Z} \rightarrow \mathbf{Y}$ of $\widehat{\mathcal{C}}$ -objects over $\mathbf{X}$ induces a morphism of abelian groups $\mathbf{M}(g): \mathbf{M}(\mathbf{Y}) \rightarrow \mathbf{M}(\mathbf{Z})$, which is compatible with the ring morphism $g^*: \mathbf{O}(\mathbf{Y}) \rightarrow \mathbf{O}(\mathbf{Z})$. In particular, when $\mathbf{Z} = \mathbf{Y}$ (g then being an $\mathbf{X}$ -endomorphism of $\mathbf{Y}$), one obtains a morphism of abelian groups

$$\mathbf{M}(g): \mathbf{M}(\mathbf{Y}) \rightarrow \mathbf{M}(\mathbf{Y})$$

which is not in general a morphism of $\mathbf{O}(\mathbf{Y})$ -modules, but which satisfies, for every $m \in \mathbf{M}(\mathbf{Y})$ and $\alpha \in \mathbf{O}(\mathbf{Y})$:

$$\mathbf{M}(g)(\alpha \cdot m) = g^*(\alpha) \cdot \mathbf{M}(g)(m).$$

This being said, one will simply write, in what follows, g^* instead of $\mathbf{M}(g)$.

Let $T \in \text{Ob}(\mathcal{C})$ and let $\mathbf{X}_T = \mathbf{X} \times T$ and $\text{pr}_{\mathbf{X}}$ be the projection $\mathbf{X}_T \rightarrow \mathbf{X}$. For every $\alpha \in \mathbf{O}(\mathbf{X}_T)$ and $g \in \mathbf{G}(T)$, set $g(\alpha) = (g^{-1})^*(\alpha)$: this is the element of $\mathbf{O}(\mathbf{X}_T)$ defined

set-theoretically by: for every $S \in \text{Ob}(\mathcal{C})$ and $x \in \mathbf{X}(S)$, $t \in T(S)$,

$$g(\alpha)(x, t) = \alpha(g^{-1}x, t).$$

One thus obtains an action (on the left) of $\mathbf{G}(T)$ by ring automorphisms on $\mathbf{O}(\mathbf{X}_T)$, functorial in T , and such that $g(\alpha) = \alpha$ if α is the image in $\mathbf{O}(\mathbf{X}_T)$ of an element of $\mathbf{O}(T)$.

Let us now denote by ϕ the identity morphism of $\mathbf{X}_T$ (cf. 6.3 and the generalization below to a morphism $\phi: \mathbf{Y} \rightarrow \mathbf{X}$) and denote $\text{Hom}_{\mathbf{X}}(\mathbf{X}_T, \mathbf{M})$ by $\mathbf{M}(\phi g^{-1})$ resp. $\mathbf{M}(\phi)$, according as $\mathbf{X}_T$ is regarded as an $\mathbf{X}$ -object via $\text{pr}_{\mathbf{X}} \circ \lambda(g^{-1})_T$, resp. $\text{pr}_{\mathbf{X}}$.

Then $\lambda(g^{-1})_T$ is an $\mathbf{X}$ -morphism between these two $\mathbf{X}$ -objects, hence, by what precedes, one obtains a morphism of abelian groups

$$(g^{-1})^*: \mathbf{M}(\phi) \rightarrow \mathbf{M}(\phi g^{-1}), \quad m \mapsto m \circ \lambda(g^{-1})_T$$

which satisfies $(g^{-1})^*(\alpha \cdot m) = (g\alpha) \cdot (g^{-1})^*(m)$ for every $m \in \mathbf{M}(\phi)$ and $\alpha \in \mathbf{O}(\mathbf{X}_T)$. (If m is a section of $\mathbf{M}_T$ over $\mathbf{X}_T$, then $(g^{-1})^*(m)$ is the section defined set-theoretically by $(x, t) \mapsto m(g^{-1}x, t)$.) In particular, $(g^{-1})^*$ is a morphism of $\mathbf{O}(T)$ -modules.

By the functoriality of the morphisms of $\mathbf{O}(\mathbf{X}_T)$ -modules $\Lambda_x(g)$, one obtains a commutative diagram:

$$\begin{array}{ccc} \mathbf{M}(\phi) & \xrightarrow{(g^{-1})^*} & \mathbf{M}(\phi g^{-1}) \\ \Lambda_{\phi}(g) \downarrow & & \downarrow \Lambda_{\phi g^{-1}}(g) \\ \mathbf{M}(g\phi) & \xrightarrow{(g^{-1})^*} & \mathbf{M}(g\phi g^{-1}) \end{array}$$

and $g\phi g^{-1} = \phi$, since ϕ is the identity map. Writing $A(g) = (g^{-1})^* \circ \Lambda_{\phi}(g)$, one therefore obtains a morphism of abelian groups

$$A(g): \mathbf{M}(\phi) \rightarrow \mathbf{M}(\phi)$$

which is “compatible with the action of $\mathbf{G}(T)$ on $\mathbf{O}(\mathbf{X}_T)$ ”, i.e. which satisfies

$$A(g)(\alpha \cdot m) = (g\alpha) \cdot A(g)(m).$$

Finally, if h is a second element of $\mathbf{G}(T)$, it follows from the functoriality of $\mathbf{M}$ and of the morphisms $\Lambda_x(g)$ that one has a commutative diagram

$$\begin{array}{ccccc} & & \mathbf{M}(\phi) & & \\ & \swarrow \Lambda_{\phi}(g) & & \searrow \Lambda_{\phi}(hg) & \\ \mathbf{M}(g\phi) & \xrightarrow{\Lambda_{\phi}(h)} & \mathbf{M}(hg\phi) & & \\ (g^{-1})^* \downarrow & & \downarrow (g^{-1})^* & \searrow ((hg)^{-1})^* & \\ \mathbf{M}(\phi) & \xrightarrow{\Lambda_{\phi}(h)} & \mathbf{M}(h\phi) & \xrightarrow{(h^{-1})^*} & \mathbf{M}(\phi) \end{array}$$

whence $A(h) \circ A(g) = A(hg)$. Consequently, one has obtained the following proposition.

Proposition 6.4.1. *For every T , the $\mathbf{O}(\mathbf{X}_T)$ -module $(\prod_{\mathbf{X}/S_0} \mathbf{M})(T) = \Gamma(\mathbf{M}_T/\mathbf{X}_T)$ is endowed, in a manner functorial in T , with an action of $\mathbf{G}(T)$ “compatible with the action of $\mathbf{G}(T)$ on $\mathbf{O}(\mathbf{X}_T)$ ”, i.e. which satisfies*

$$A(g)(\alpha \cdot m) = g(\alpha) \cdot A(g)(m).$$

As $g(\alpha) = \alpha$ for $\alpha \in \mathbf{O}(T)$, this shows, in particular, that $\prod_{\mathbf{X}/S_0} \mathbf{M}$ is a $\mathbf{G}\text{-}\mathbf{O}_{S_0}$ -module.

More generally let, as in 6.3, $\mathbf{Y}$ be a second G -equivariant $\hat{\mathcal{C}}$ -object, $\phi: \mathbf{Y} \rightarrow \mathbf{X}$ a $\hat{\mathcal{C}}$ -morphism and $\mathbf{H} = \text{Stab}_{\mathbf{G}}(\phi)$. Then the fiber product $\mathbf{M}_{\phi} = \mathbf{Y} \times_{\phi} \mathbf{M}$ is an $\mathbf{H}$ -equivariant $\mathbf{O}_{\mathbf{Y}}$ -module. One therefore obtains the:

Corollary 6.4.2. *The functor $\prod_{\mathbf{Y}/S_0} \mathbf{M}_{\phi}$ is a $\text{Stab}_{\mathbf{G}}(\phi)\text{-}\mathbf{O}_{S_0}$ -module.*

6.5. G -equivariant $\mathcal{O}_X$ -modules. — Let us apply what precedes in the following situation. Let S be a scheme, G an S -group scheme acting on an S -scheme X , and $\mathcal{F}$ an $\mathcal{O}_X$ -module (not necessarily quasi-coherent).

Definition 6.5.1. One says that $\mathcal{F}$ is a G -equivariant $\mathcal{O}_X$ -module if the $\mathbf{O}_X$ -module $\mathbf{M} = \mathbf{W}(\mathcal{F})$ is G -equivariant, i.e. if one has given oneself, for every morphism $(g, x): U \rightarrow G \times_S X$, isomorphisms of $\mathbf{O}_U$ -modules

$$\Lambda_x(g): \mathbf{W}(x^*(\mathcal{F})) \xrightarrow{\sim} \mathbf{W}((gx)^*(\mathcal{F})),$$

functorial in U and satisfying $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$. As the functor $\mathbf{W}$ is fully faithful (cf. 4.6.1.1 and 4.6.2), one therefore obtains isomorphisms of $\mathcal{O}_U$ -modules

$$\Lambda_x(g): x^*(\mathcal{F}) \xrightarrow{\sim} (gx)^*(\mathcal{F}),$$

where one recalls that gx denotes the morphism $\lambda \circ (g, x): U \rightarrow X$. In particular, applying this to the identity morphism of $G \times_S X$, one obtains an isomorphism of $\mathcal{O}_{G \times_S X}$ -modules

$$(\star) \quad \theta: \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

such that the diagram $(\star\star)$ of morphisms of sheaves on $G \times_S G \times_S X$ below is commutative:

$$\begin{array}{ccc}
 \mathrm{pr}_{2,3}^* \circ \mathrm{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{\mathrm{pr}_{2,3}^*(\theta)} & \mathrm{pr}_{2,3}^* \circ \lambda^*(\mathcal{F}) & (\mathrm{id}_G \times \lambda)^* \circ \mathrm{pr}_X^*(\mathcal{F}) \\
 & & (\star\star) & \downarrow (\mathrm{id}_G \times \lambda)^*(\theta) \\
 (\mu \times \mathrm{id}_X)^* \circ \mathrm{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{(\mu \times \mathrm{id}_X)^*(\theta)} & (\mu \times \mathrm{id}_X)^* \circ \lambda^*(\mathcal{F}) & (\mathrm{id}_G \times \lambda)^* \circ \lambda^*(\mathcal{F})
 \end{array}$$

Moreover, if $\mathcal{E}$ is a second G -equivariant $\mathcal{O}_X$ -module, one says that a morphism of $\mathcal{O}_X$ -modules $\phi: \mathcal{E} \rightarrow \mathcal{F}$ is G -equivariant if the morphism $\mathbf{W}(\phi): \mathbf{W}(\mathcal{E}) \rightarrow \mathbf{W}(\mathcal{F})$ is. With the notations above, this amounts to saying that $\theta_{\mathcal{F}} \circ \mathrm{pr}_X^*(\phi) = \lambda^*(\phi) \circ \theta_{\mathcal{E}}$.

Remark 6.5.2. If $f: Y \rightarrow X$ is a G -equivariant morphism, it follows from 6.3.1 that $f^*(\mathcal{F})$ is a G -equivariant $\mathcal{O}_Y$ -module.

Remark 6.5.3. Suppose moreover that $\mathcal{F}$ is *quasi-coherent*. Then it follows from what precedes that the datum of a structure of G -equivariant $\mathbf{O}_X$ -module on $\mathbf{M} = \mathbf{W}(\mathcal{F})$ amounts to the datum of such a structure on $\mathbf{V}(\mathcal{F})$, which still amounts to giving oneself an action of G on the vector fibration $\mathbf{V}(\mathcal{F})$, compatible with the action on X and “linear” on the fibers.

Indeed, denote by ϕ the isomorphism $\lambda^*(\mathcal{F}) \xrightarrow{\sim} \mathrm{pr}_X^*(\mathcal{F})$ inverse to θ . For every morphism $(g, x): U \rightarrow G \times_S X$, one has an isomorphism of $\mathcal{O}_U$ -modules $\phi_x(g) = \Lambda_x(g)^{-1} = \Lambda_{gx}(g^{-1})$ from $(gx)^*(\mathcal{F})$ to $x^*(\mathcal{F})$; it induces an isomorphism of $\mathbf{O}_U$ -modules

$$\mathbf{V}((gx)^*(\mathcal{F})) \xrightarrow{\sim} \mathbf{V}(x^*(\mathcal{F}))$$

which one will denote by ${}^t\phi_x(g)$. One then has

$${}^t\phi_{gx}(h) \circ {}^t\phi_x(g) = {}^t(\Lambda_{gx}(h) \circ \phi_x(g))^{-1} = {}^t\Lambda_x(hg)^{-1} = {}^t\phi_x(hg).$$

As, for every X -scheme $f: Y \rightarrow X$, one has $\mathbf{V}(\mathcal{F}) \times_f Y \simeq \mathbf{V}(f^*(\mathcal{F}))$ (cf. 4.6.2), one therefore obtains that the isomorphism

$$\begin{aligned}
 {}^t\phi: G \times_S \mathbf{V}(\mathcal{F}) &= (G \times_S X) \times_{\mathrm{pr}_X} \mathbf{V}(\mathcal{F}) = \mathbf{V}(\mathrm{pr}_X^*(\mathcal{F})) \\
 &\xrightarrow{\sim} \mathbf{V}(\lambda^*(\mathcal{F})) = (G \times_S X) \times_{\lambda} \mathbf{V}(\mathcal{F})
 \end{aligned}$$

endows $\mathbf{V}(\mathcal{F})$ with a structure of G -equivariant $\mathbf{O}_X$ -module. Finally, identifying each functor $\mathbf{V}(-)$ with the vector fibration which represents it, and composing ${}^t\phi$ with the projection onto $\mathbf{V}(\mathcal{F})$, one obtains an action of G on the vector fibration $\mathbf{V}(\mathcal{F})$, compatible with the action on X and “linear” on the fibers.

One thus recovers the definition given, for example, in [GIT, Chap. 1, § 3], except that in *loc. cit.* Mumford considers an $\mathcal{O}_X$ -module $\mathcal{E}$ locally free of finite rank, and defines an

action of G on $\mathbf{V}(\mathcal{E}) = \mathbf{W}(\mathcal{E}^\vee)$. Indeed, the preceding diagram $(\star\star)$, with θ replaced by ϕ and the direction of the arrows reversed, is exactly the one found in *loc. cit.*, Def. 1.6, and the isomorphism ${}^t\phi$ above coincides with the isomorphism Φ of *loc. cit.*, p. 31.

Remark 6.5.4. Let us consider in particular the case where $X = S$, endowed with the trivial action of G . In this case, a G -equivariant $\mathcal{O}_S$ -module $\mathcal{F}$ is the same thing as a G - $\mathbf{O}_S$ -module (cf. 4.7.1). Moreover, if one denotes by f the morphism $G \rightarrow S$ (here equal both to pr_X and to λ), then the isomorphism

$$(\star) \quad \theta: f^*(\mathcal{F}) \xrightarrow{\sim} f^*(\mathcal{F})$$

is an element of

$$\begin{aligned} \text{Hom}_{\mathcal{O}_G}(f^*(\mathcal{F}), f^*(\mathcal{F})) &= \text{Hom}_{\mathbf{O}_G}(\mathbf{W}(f^*(\mathcal{F})), \mathbf{W}(f^*(\mathcal{F}))) \\ &= \text{End}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}))(G) \end{aligned}$$

which is none other than the morphism $\rho: G \rightarrow \text{End}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}))$ defining the operation of G on $\mathbf{W}(\mathcal{F})$. Moreover, θ corresponds by adjunction to the morphism of $\mathcal{O}_S$ -modules $f_*(\theta) \circ \tau: \mathcal{F} \rightarrow f_*f^*(\mathcal{F})$, where $\tau: \mathcal{F} \rightarrow f_*f^*(\mathcal{F})$ is the “unit” morphism of the adjunction. (This will be used in VI_B, 11.10.bis.)

6.6. The functors $\prod_{X/S} \mathbf{W}(\mathcal{F})$ and $\mathbf{W}(p_*(\mathcal{F}))$. — Let S be a scheme, G an S -group scheme acting on S -schemes X and Y via the morphisms $\lambda: G \times_S X \rightarrow X$ and $\mu: G \times_S Y \rightarrow Y$, and let $p: X \rightarrow Y$ be a G -equivariant morphism. Suppose that the morphisms $p: X \rightarrow Y$ and $\nu_Y: Y \rightarrow S$ are *quasi-compact* and *quasi-separated*, and that $\pi: G \rightarrow S$ is *flat*.

Then the projection $\text{pr}_Y: G \times_S Y \rightarrow Y$ is flat, as is μ (since μ is the composite of pr_Y and of the automorphism $(g, y) \mapsto (g, gy)$). Finally, for a variable S -scheme $f: T \rightarrow S$, one will denote by $p^T: X_T \rightarrow T$ and $f_X: X_T \rightarrow X$ the morphisms deduced from p and f by base change.

Let $\mathcal{F}$ be a *quasi-coherent* and G -equivariant $\mathcal{O}_X$ -module, and let θ be the isomorphism $\text{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$ of 6.5.1 $(\star)$. As p is quasi-compact and quasi-separated, $p_*(\mathcal{F})$ is quasi-coherent (cf. EGA I, 9.2.1).

Lemma 6.6.1. *The quasi-coherent $\mathcal{O}_Y$ -module $p_*(\mathcal{F})$ is G -equivariant.*

Indeed, one has the two cartesian squares below:

$$\begin{array}{ccccc} G \times_S X & \xrightarrow{\text{pr}_X} & X & \xleftarrow{\lambda} & G \times_S X \\ q \downarrow & & \downarrow p & & \downarrow q \\ G \times_S Y & \xrightarrow{\text{pr}_Y} & Y & \xleftarrow{\mu} & G \times_S Y. \end{array}$$

As p is quasi-compact and quasi-separated and pr_Y and μ are flat, it follows from EGA III, 1.4.15 (supplemented by EGA IV₁, 1.7.21) that one has $q^*\text{pr}_X^*(\mathcal{F}) = \text{pr}_Y^*p_*(\mathcal{F})$ and $q^*\lambda^*(\mathcal{F}) = \mu^*p_*(\mathcal{F})$. Consequently, θ induces an isomorphism

$$\theta_Y: \text{pr}_Y^*p_*(\mathcal{F}) \xrightarrow{\sim} \mu^*p_*(\mathcal{F}),$$

and one likewise obtains that θ_Y satisfies the “cocycle condition” 6.5.1 $(\star\star)$ (since the morphisms $G \times_S G \times_S Y \rightarrow G \times_S Y$ which occur are flat). This proves the lemma.

In particular, take $Y = S$ endowed with the trivial action of G . Taking remark 6.5.4 into account, one then obtains that $p_*(\mathcal{F})$ is a G - $\mathbf{O}_S$ -module. If moreover G is affine over S and if one writes $\mathcal{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ is therefore an $\mathcal{A}(G)$ -comodule, by 4.7.2.

On the other hand, by 6.4.1, the functor $\prod_{X/S} \mathbf{W}(\mathcal{F})$, which to every $f: T \rightarrow S$ associates

$$\mathbf{W}(f_X^*(\mathcal{F}))(X_T) = \Gamma(X_T, f_X^*(\mathcal{F})) = \Gamma(T, p_X^T f_X^*(\mathcal{F}))$$

is a $G\text{-}\mathbf{O}_S$ -module. Moreover, one has a canonical morphism $\tau: \mathbf{W}(p_*(\mathcal{F})) \rightarrow \prod_{X/S} \mathbf{W}(\mathcal{F})$, which is given for every $f: T \rightarrow S$ by the canonical morphism

$$\Gamma(T, f^*p_*(\mathcal{F})) \rightarrow \Gamma(T, p_*^T f_X^*(\mathcal{F}))$$

and which is an isomorphism when one restricts it to the full subcategory of schemes T flat over S , and one checks without difficulty that τ is a morphism of $G\text{-}\mathbf{O}_S$ -modules. Consequently, one has obtained the following proposition (for point (ii), compare with [GIT], p. 32).

Proposition 6.6.2. *Let S be a scheme, G an S -group scheme acting on an S -scheme X , and $\mathcal{F}$ a quasi-coherent and G -equivariant $\mathcal{O}_X$ -module. Suppose that $\pi: G \rightarrow S$ is flat and that $p: X \rightarrow S$ is quasi-compact and quasi-separated.*

- (i) *Then $p_*(\mathcal{F})$ is a quasi-coherent $G\text{-}\mathbf{O}_S$ -module. On the other hand, the canonical morphism $\mathbf{W}(p_*(\mathcal{F})) \rightarrow \prod_{X/S} \mathbf{W}(\mathcal{F})$ is a morphism of $G\text{-}\mathbf{O}_S$ -modules, and these two functors coincide on the category of flat S -schemes.*
- (ii) *If moreover G is affine over S and if one writes $\mathcal{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ is endowed with a structure of $\mathcal{A}(G)$ -comodule.⁽⁵⁵⁾*

6.7. Stabilizers. — Let S be a scheme, G an S -group scheme acting on an S -scheme X , and let $\mathcal{F}$ be a quasi-coherent G -equivariant $\mathcal{O}_X$ -module. Let Y be a second S -scheme endowed with an action of G (possibly trivial), let $\phi: Y \rightarrow X$ be an S -morphism, not necessarily G -equivariant, and let $\text{Stab}_G(\phi)$ be the stabilizer in G of ϕ (cf. 6.3.2).

Let us note at once (see Exp. VI_B, § 6) that $\text{Stab}_G(\phi)$ is representable by a closed group subscheme H of G if X is separated over S and if Y is essentially free over S (cf. *loc. cit.*, Def. 6.2.1). Indeed, consider the morphism $r: G \times_S Y \rightarrow X \times_S X$ given set-theoretically by $r(g, y) = (\phi(y), g\phi(g^{-1}y))$, and let $P = G \times_S Y$ and P' be the inverse image by r of the diagonal $\Delta_{X/S}$. Then one has (cf. *loc. cit.*, 6.2.4 (a))

$$\text{Stab}_G(\phi) = \prod_{P/G} P'$$

and hence, by *loc. cit.*, $\text{Stab}_G(\phi)$ is representable by a closed group subscheme H of G if X is separated over S and if Y is essentially free over S ; this second condition being automatically satisfied if S is the spectrum of a field, or else if $Y = S$. Under these hypotheses, $\phi^*(\mathcal{F})$ is then a quasi-coherent H -equivariant $\mathcal{O}_Y$ -module (cf. 6.5.2).

Therefore, if moreover $\pi: G \rightarrow S$ and $p: Y \rightarrow S$ are quasi-compact and quasi-separated over S , and π is flat, then $p_*\phi^*(\mathcal{F})$ is an $H\text{-}\mathbf{O}_S$ -module, by 6.6.2. In particular, one obtains the:

Corollary 6.7.1. *Let S be a scheme, G an S -group scheme acting on an S -scheme X , and $\mathcal{F}$ a quasi-coherent G -equivariant $\mathcal{O}_X$ -module. Suppose that $\pi: G \rightarrow S$ is flat and that $X \rightarrow S$ is quasi-compact and separated. Let $\tau: S \rightarrow X$ be a section of X over S .*

- (i) *The stabilizer $H = \text{Stab}_G(\tau)$ is a closed group subscheme of G , and $\tau^*(\mathcal{F})$ is a quasi-coherent $H\text{-}\mathbf{O}_S$ -module.*
- (ii) *If moreover H is affine over S (for example, if G is) and if one writes $\mathcal{A}(H) = \pi_*(\mathcal{O}_H)$, then $\tau^*(\mathcal{F})$ is an $\mathcal{A}(H)$ -comodule.*

6.8. G -equivariant sheaves on G . — To conclude, let us indicate two results (6.8.1 and 6.8.6 below) which will be used in exposés II and III (cf. in particular III, 4.25).

Proposition 6.8.1. *Let S be a scheme, $\pi: G \rightarrow S$ an S -group scheme, and $\varepsilon: S \rightarrow G$ the unit section. One considers the action by left translations of G on itself. Then the functors $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ and $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ induce equivalences, quasi-inverse to one another, between the category of quasi-coherent $\mathcal{O}_S$ -modules and that of quasi-coherent G -equivariant $\mathcal{O}_G$ -modules.*

⁽⁵⁵⁾**N.D.E.** The same holds if G is flat, quasi-compact and quasi-separated over S and if $p_*(\mathcal{F})$ is a flat $\mathcal{O}_S$ -module, cf. Exp. VI_B, 11.6.1 (ii).

PROOF. Denote by μ the multiplication of G and by pr_2 the second projection $G \times_S G \rightarrow G$. As $\pi \circ \mu = \pi \circ \text{pr}_2$, for every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$ one has a canonical isomorphism $\mu^* \pi^*(\mathcal{E}) = \text{pr}_2^* \pi^*(\mathcal{E})$, and one easily checks that this isomorphism satisfies the “cocycle condition” 6.5.1(★), i.e. $\pi^*(\mathcal{E})$ is a G -equivariant $\mathcal{O}_G$ -module. As $\varepsilon^* \pi^*(\mathcal{E}) = \mathcal{E}$, the functor $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ is fully faithful; it therefore remains to see that for every G -equivariant quasi-coherent $\mathcal{O}_G$ -module $\mathcal{F}$, one has $\mathcal{F} \simeq \pi^* \varepsilon^*(\mathcal{F})$.

By hypothesis, one has an isomorphism $\theta: \text{pr}_2^*(\mathcal{F}) \xrightarrow{\sim} \mu^*(\mathcal{F})$; taking the inverse image of θ by the morphism $\tau: G \rightarrow G \times_S G$ with components $(\text{id}_G, \varepsilon \circ \pi)$, one obtains an isomorphism $\mathcal{F} \xrightarrow{\sim} \pi^* \varepsilon^*(\mathcal{F})$.

Remarks 6.8.2. (a) Let us consider the action of $G \times_S G$ on G defined by $(g_1, g_2) \cdot g = g_1 g g_2^{-1}$; then the stabilizer of the unit section $\varepsilon: S \rightarrow G$ is the diagonal subgroup H of $G \times_S G$. Consequently, if $\mathcal{F}$ is a quasi-coherent $(G \times_S G)$ -equivariant $\mathcal{O}_G$ -module, then, by 6.5.2, $\mathcal{E} = \varepsilon^*(\mathcal{F})$ is endowed with the structure of an H - $\mathcal{O}_S$ -module.

(b) One can show that $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ is an equivalence of categories, between the category of quasi-coherent $(G \times_S G)$ -equivariant $\mathcal{O}_G$ -modules and that of quasi-coherent H - $\mathcal{O}_S$ -modules. This is a particular case of more general “descent” results (cf. Exp. IV, §2 and SGA 1, VIII); see for example [Th87], 1.2–1.3.

Remark 6.8.3. One keeps the notations of 6.8.1. For every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$, let us denote by $\pi_*^G \pi^*(\mathcal{E})$ the sub- $\mathcal{O}_S$ -module of $\pi_* \pi^*(\mathcal{E})$ whose sections over every open V of S are the $\gamma \in \Gamma(\pi^{-1}(V), \pi^*(\mathcal{E}))$ such that $g \cdot \gamma_{S'} = \gamma_{S'}$ for every $S' \rightarrow V$ and every $g \in G(S')$. Then the natural morphism $\mathcal{E} \rightarrow \pi_*^G \pi^*(\mathcal{E})$ is an isomorphism: this is immediate if $\pi_* \pi^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_S} \pi_*(\mathcal{O}_G)$ (for example if $G \rightarrow S$ is affine, or if $G \rightarrow S$ is quasi-compact and quasi-separated and $\mathcal{E}$ is flat), and this is verified without difficulty in the general case.

Remark 6.8.4. Let S be a scheme, H an S -group scheme operating on an S -scheme X , and $\mathcal{F}$ an $\mathcal{O}_X$ -module. One assumes that H is flat over S , and one denotes by $\mathbf{W}_P(\mathcal{F})$ the restriction of the functor $\mathbf{W}(\mathcal{F})$ to the full subcategory consisting of the flat S -schemes. Since, by 6.5.1, endowing $\mathcal{F}$ with an H -equivariant module structure is equivalent to giving an isomorphism $\theta: \text{pr}_X^*(\mathcal{F}) \rightarrow \lambda^*(\mathcal{F})$ of sheaves on $G \times_S X$, satisfying the “cocycle condition” (★), one sees that, in order to give an H -equivariant module structure on $\mathcal{F}$, it suffices to give such a structure on $\mathbf{W}_P(\mathcal{F})$.

Recall 6.8.5. Let X be an S -scheme and Y a sub- S -scheme of X . One denotes by $\mathcal{N}_{Y/X}$ the conormal sheaf of the immersion $i: Y \hookrightarrow X$ (cf. EGA IV₄, 16.1.2). If $S' \rightarrow S$ is a flat morphism and if one denotes by $i': Y' \hookrightarrow X'$ the immersion deduced from i by base change, then, by loc. cit., 16.2.2 (iii), one has $\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'} = \mathcal{N}_{Y/X}$.

Proposition 6.8.6. Let S be a scheme, X an S -group scheme, and Y a group sub- S -scheme of X . One assumes that Y is flat over S . Then the conormal sheaf $\mathcal{N}_{Y/X}$ is a $(Y \times_S Y)$ -equivariant $\mathcal{O}_Y$ -module.

Indeed, $H = Y \times_S Y$ is flat over S , so, by remark 6.8.4, it suffices to endow $\mathbf{W}_P(\mathcal{N}_{Y/X})$ with the structure of an H -equivariant module. Let S' be a flat S -scheme, let $Y' \hookrightarrow X'$ be the immersion obtained by base change, and let $\mathcal{N}' = \mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'}$. By 6.8.5, one has $\mathcal{N}' = \mathcal{N}_{Y'/X'}$.

Every $h \in H(S')$ induces an automorphism of Y' , and one therefore obtains, for every $y \in Y'(S')$, isomorphisms

$$\Gamma(S', y^*(\mathcal{N}')) \xrightarrow{\sim} \Gamma(S', y^* h^*(\mathcal{N}'))$$

which endow $\mathbf{W}_P(\mathcal{N})$ with the structure of an H -equivariant module (cf. 6.1).

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